CHAPTER 4

JACOBIAN ANALYSIS OF SERIAL MANIPULATORS

4.1 INTRODUCTION

In previous chapters we have studied the kinematic relations between the end-effector location and the joint variables of serial and parallel manipulators. Both the direct and inverse kinematics have been analyzed. This knowledge enables us to bring the end effector to some desired locations in space. In this chapter we extend our study from a position analysis problem to a velocity analysis problem of serial manipulators.

For some applications, such as spray painting (Fig. 4.1), it is necessary to move the end effector of a manipulator along some desired paths with a prescribed speed. To achieve this goal, the motion of the individual joints of a manipulator must be carefully coordinated. There are two types of velocity coordination problems, called direct velocity and inverse velocity problems. For the *direct velocity problem*, the input joint rates are given and the objective is to find the velocity state of the end effector. For the *inverse velocity problem*, the velocity state of the end effector is given and the input joint rates required to produce the desired velocity are to be found. In this chapter, the fundamental knowledge needed to achieve such a coordinated motion is developed.

We call the vector space spanned by the joint variables the *joint space*, and the vector space spanned by the end-effector location, the *end-effector space*. For robot manipulators, the *Jacobian matrix*, or simply *Jacobian*, is defined as the matrix that transforms the joint rates in the actuator space to the velocity state in the end-effector space.

FIGURE 4.1. Spray painting robot. (Courtesy of Fanuc Robotics North America, Inc., Rochester Hills, Michigan.)

The Jacobian matrix is a critical component for generating trajectories of prescribed geometry in the end-effector space. Most coordination algorithms employed by industrial robots avoid numerical inversion of the Jacobian matrix by deriving analytical inverse solutions on an ad hoc basis. Therefore, it is important that efficient algorithms be developed. Since the velocity state of the end effector can be defined in various ways, a variety of Jacobian matrices and consequently, different methods of formulation have appeared in the literature (Craig, 1986; Featherstone, 1983; Hollerbach and Saher, 1983; Hunt, 1986, 1987a,b; Orin and Schrader, 1984; Waldron et al., 1985; Whitney, 1972). In what follows, two different definitions of the Jacobian matrix are described. The first is a *conventional Jacobian* and the second is a *screw-based Jacobian.*

The Jacobian matrix is also useful in other applications. For some configurations of a manipulator, the Jacobian matrix may lose its full rank. Such conditions are called *singular conditions*. At a singular condition, a serial manipulator may lose one or more degrees of freedom while a parallel manipulator may gain one or more degrees of freedom. In this chapter, the singular conditions of serial manipulators are also studied.

4.2 DIFFERENTIAL KINEMATICS OF A RIGID BODY

We first study the differential kinematics of a rigid body. Then these kinematic properties are applied for a derivation of the differential kinematics of the links in a manipulator and for a development of the Jacobian matrix. Since we will be dealing with many frames of reference, the following notations are made to identify the frame with respect to which a vector is defined. A vector $\mathbf{p}$ can be a function of time in one reference frame but constant in another reference frame. Thus, in general, we need two frames of references to describe the nature of a vector: one with respect to which the change of a vector is measured and another in which the vector is expressed. In this book we use an inner leading superscript to denote the frame with respect to which a vector is being measured, and an outer leading superscript to indicate the frame in which the vector is expressed.

For example, ${}^{B}\mathbf{p}$ denotes the position vector of a point P with respect to frame B, and ${}^{A}({}^{B}\mathbf{p})$ denotes ${}^{B}\mathbf{p}$ expressed in frame A. Similarly, the velocity of P is defined by taking the derivative of ${}^{B}\mathbf{p}$ with respect to time:

$$ {}^{B}\mathbf{v}_{p} = \frac{d\,{}^{B}\mathbf{p}}{dt}. \tag{4.1} $$

However, once the differentiation is taken, the vector can be expressed in any other frame. Thus

$$ {}^{A}({}^{B}\mathbf{v}_{p}) = {}^{A}\left(\frac{d\,{}^{B}\mathbf{p}}{dt}\right) \tag{4.2} $$

indicates that the differentiation is taken with respect to frame B and the resulting vector is expressed in frame A.

When the two leading superscripts are the same or when the frame with respect to which a vector quantity is being measured is clearly understood, the inner superscript will be omitted. For clarity, we often use the rotation matrix ${}^{A}R_{B}$ to transform a vector from one reference frame to another:

$$ {}^{A}({}^{B}\mathbf{p}) \equiv {}^{A}R_{B}\,{}^{B}\mathbf{p}. \tag{4.3} $$

Furthermore, when no specific reference frame is mentioned, either the base frame is implied or any reference frame can be used. Note that all vectors in one equation must be expressed in the same reference frame.

4.2.1 Angular Velocity of a Rigid Body

While the linear velocity describes the rate of change of the position of a point in space, the angular velocity vector describes the rate of change of the

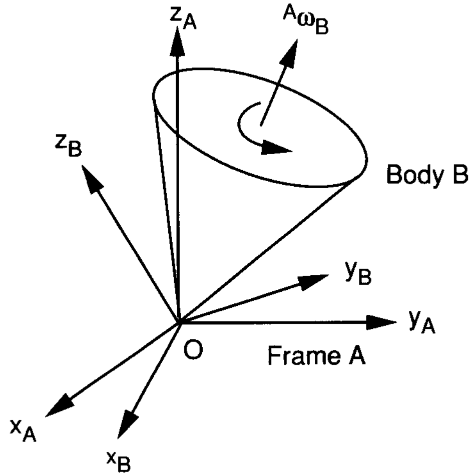

FIGURE 4.2. Instantaneous rotation of frame B with respect to A.

orientation of a rigid body. Figure 4.2 shows that frame B is rotating with respect to frame A with a fixed point O. The orientation of frame B with respect to A can be described by a rotation matrix, ${}^AR_B$. Since the rotation matrix ${}^AR_B$ is orthogonal, the inverse transformation of ${}^AR_B$ is identical to the transpose. Hence

$$ {}^AR_B\, {}^AR_B^{\mathrm{T}} = I, \tag{4.4}$$

where I is a 3×3 identity matrix.

Taking the derivative of Eq. (4.4) with respect to time, we obtain

$$ {}^A\dot{R}_B\, {}^AR_B^{\mathrm{T}} + {}^AR_B\, {}^A\dot{R}_B^{\mathrm{T}} = 0. \tag{4.5}$$

Substituting ${}^AR_B^{\mathrm{T}} = {}^AR_B^{-1}$ and ${}^AR_B = ({}^AR_B^{-1})^{\mathrm{T}}$ into Eq. (4.5), we obtain

$$ ({}^A\dot{R}_B\, {}^AR_B^{-1}) + ({}^A\dot{R}_B\, {}^AR_B^{-1})^{\mathrm{T}} = 0. \tag{4.6}$$

Hence ${}^A\dot{R}_B\, {}^AR_B^{-1}$ is a 3×3 skew-symmetric matrix. Without losing generality, we may define the skew-symmetric matrix as

$$ \Omega \equiv {}^A\dot{R}_B\, {}^AR_B^{-1} = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}. \tag{4.7}$$

Here ω_x, ω_y, and ω_z are to be identified as three independent parameters specifying the angular velocity of a rigid body. In what follows it will be shown that these three quantities form the components of a vector called the angular velocity vector of B in A.

The position vector of a point P that is embedded in frame B and measured with respect to frame A is given by

$$ {}^{A}\mathbf{p} = {}^{A}R_{B}\, {}^{B}\mathbf{p}. \tag{4.8} $$

Note that ${}^{B}\mathbf{p}$ is a constant vector in frame B since P is embedded in B. The velocity of P with respect to frame A is obtained by taking the derivative of Eq. (4.8) with respect to time:

$$ {}^{A}\mathbf{v}_{p} = \frac{d}{dt}({}^{A}R_{B}\, {}^{B}\mathbf{p}) = {}^{A}\dot{R}_{B}\, {}^{B}\mathbf{p}. \tag{4.9} $$

Solving ${}^{B}\mathbf{p}$ from Eq. (4.8) and substituting the resulting expression into Eq. (4.9) yields

$$ {}^{A}\mathbf{v}_{p} = {}^{A}\dot{R}_{B}\, {}^{A}R_{B}^{-1}\, {}^{A}\mathbf{p}. \tag{4.10} $$

Substituting Eq. (4.7) into (4.10) produces

$$ {}^{A}\mathbf{v}_{p} = \Omega\, {}^{A}\mathbf{p}. \tag{4.11} $$

We may ask ourselves the following question: Is there any point in B that has zero velocity at that instant? Assuming that $\tilde{P}$ is such a point,

$$ {}^{A}\mathbf{v}_{\tilde{p}} = \Omega\, {}^{A}\tilde{\mathbf{p}} = \mathbf{0}. \tag{4.12} $$

Equation (4.12) consists of three homogeneous linear equations in three unknowns, $\tilde{p}_x$, $\tilde{p}_y$, and $\tilde{p}_z$. The compatibility condition for the existence of nontrivial solutions is that the determinant of the coefficient matrix must vanish (i.e., $|\Omega| = 0$). Since Ω is a 3×3 skew-symmetric matrix, this condition is satisfied automatically:

$$ \begin{vmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{vmatrix} = \omega_x\omega_y\omega_z - \omega_x\omega_y\omega_z = 0. $$

Hence only two of the three equations in Eq. (4.12) are independent. Solving Eq. (4.12) for the ratio $\tilde{p}_x : \tilde{p}_y : \tilde{p}_z$, we obtain

$$ \tilde{p}_x : \tilde{p}_y : \tilde{p}_z = \omega_x : \omega_y : \omega_z. \tag{4.13} $$

We conclude that there exist infinitely many stationary points, and these points lie on a line that passes through the origin and is parallel to the vector

${}^{A}\boldsymbol{\omega}_B = [\omega_x, \omega_y, \omega_z]^{\mathrm{T}}$. We call the vector ${}^{A}\boldsymbol{\omega}_B$ the *angular velocity vector* and the line the *instantaneous screw axis*. Using the vector notation, Eq. (4.11) can be written as

$${}^{A}\mathbf{v}_p = {}^{A}\boldsymbol{\omega}_B \times {}^{A}\mathbf{p}. \tag{4.14}$$

4.2.2 Linear Velocity of a Point

Figure 4.3 shows a rigid body B that is making an instantaneous rotation as well as translation with respect to a reference frame A. The position vector of a point P, which is not necessarily fixed in frame B, relative to frame A can be written as

$${}^{A}\mathbf{p} = {}^{A}\mathbf{q} + {}^{A}R_B\, {}^{B}\mathbf{p}, \tag{4.15}$$

where ${}^{A}\mathbf{q} = \overline{OQ}$ denotes the position vector of the origin Q of frame B with respect to frame A.

To derive the velocity of P, we first consider the rate of change of the second term in Eq. (4.15). This is essentially the case when frame B is rotating with respect to frame A with the origin Q fixed in A. Differentiating the second term of Eq. (4.15) with respect to time yields

$$\frac{d}{dt}({}^{A}R_B\, {}^{B}\mathbf{p}) = {}^{A}R_B\, {}^{B}\mathbf{v}_p + {}^{A}\dot{R}_B\, {}^{B}\mathbf{p}, \tag{4.16}$$

where ${}^{B}\mathbf{v}_p = \frac{d}{dt}\, {}^{B}\mathbf{p}$ denotes the velocity of P with respect frame B.

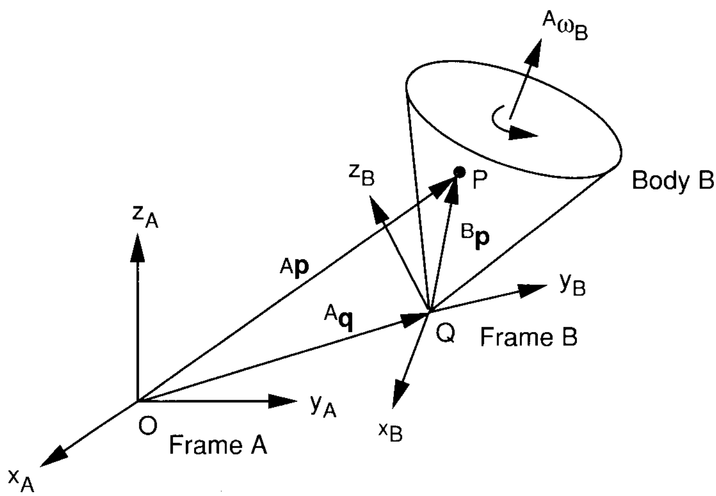

FIGURE 4.3. Instantaneous motion of a rigid body B with respect to frame A.

Postmultiplying both sides of Eq. (4.7) by ${}^A R_B$, we obtain

$$ {}^A\dot{R}_B = \Omega\, {}^A R_B. \tag{4.17} $$

Substituting Eq. (4.17) into (4.16) yields

$$ \frac{d}{dt}({}^A R_B\, {}^B\mathbf{p}) = {}^A R_B\, {}^B\mathbf{v}_p + \Omega\, {}^A R_B\, {}^B\mathbf{p}. \tag{4.18} $$

Equation (4.18) can be written in vector form as

$$ \frac{d}{dt}({}^A R_B\, {}^B\mathbf{p}) = {}^A R_B\, {}^B\mathbf{v}_p + {}^A\boldsymbol{\omega}_B \times ({}^A R_B\, {}^B\mathbf{p}). \tag{4.19} $$

When the origin Q of frame B is moving with respect to frame A, we simply add a component representing the linear velocity of Q in A to Eq. (4.19). Hence a general equation of motion can be written as

$$ {}^A\mathbf{v}_p = {}^A\mathbf{v}_q + {}^A R_B\, {}^B\mathbf{v}_p + {}^A\boldsymbol{\omega}_B \times ({}^A R_B\, {}^B\mathbf{p}), \tag{4.20} $$

where ${}^A\mathbf{v}_q = {}^A\dot{q}$ denotes the velocity of Q relative to frame A. The first term in Eq. (4.20) is contributed by the linear velocity of Q with respect to frame A, the second term is contributed by the relative motion of P with respect to frame B, and the third term is contributed by the rotation of frame B with respect to A.

Special Case. If point P is embedded in the moving frame B, ${}^B\mathbf{v}_p = 0$ identically. Equation (4.20) reduces to

$$ {}^A\mathbf{v}_p = {}^A\mathbf{v}_q + {}^A\boldsymbol{\omega}_B \times ({}^A R_B\, {}^B\mathbf{p}). \tag{4.21} $$

Although Eq. (4.21) is derived for the case in which Q is the origin of a moving frame, it is equally applicable to any two points fixed on the moving frame. In general, if P and Q are two points embedded in a rigid body B, their velocities are related by the equation

$$ {}^A\mathbf{v}_p = {}^A\mathbf{v}_q + {}^A\boldsymbol{\omega}_B \times ({}^A\mathbf{p} - {}^A\mathbf{q}). \tag{4.22} $$

4.2.3 Instantaneous Screw Axis

In this section we show that a general instantaneous motion of a rigid body can be described by a differential rotation about a unique axis and a differential translation along the same axis. This concept will be applied to the Jacobian analysis of serial manipulators.

For a general spatial motion of a rigid body B, are there any stationary points in B? If ${}^B\tilde{\mathbf{p}}$ is a stationary point, ${}^A\mathbf{v}_{\tilde{p}} = 0$ identically, and Eq. (4.21) reduces to

$$ {}^{A}\boldsymbol{\omega}_{B} \times ({}^{A}R_{B}\, {}^{B}\tilde{\mathbf{p}}) = -{}^{A}\mathbf{v}_{q}. \tag{4.23}$$

Since the angular velocity ${}^{A}\boldsymbol{\omega}_{B}$ is derived from a 3×3 skew-symmetric matrix Ω, the coefficients matrix of Eq. (4.23) is singular. It follows that, in general, there are no solutions to Eq. (4.21). However, we may seek for those points whose linear velocity vectors point along the direction of the angular velocity. That is,

$$ {}^{A}\mathbf{v}_{\tilde{p}} = \lambda {}^{A}\boldsymbol{\omega}_{B}, \tag{4.24}$$

where λ is called a *pitch*.

Substituting Eq. (4.24) into (4.21) yields

$$ {}^{A}\mathbf{v}_{q} + {}^{A}\boldsymbol{\omega}_{B} \times ({}^{A}R_{B}\, {}^{B}\tilde{\mathbf{p}}) = \lambda\, {}^{A}\boldsymbol{\omega}_{B}. \tag{4.25}$$

Dot-multiplying both sides of Eq. (4.25) by ${}^{A}\boldsymbol{\omega}_{B}$, we obtain

$$ \lambda = \frac{{}^{A}\boldsymbol{\omega}_{B} \cdot {}^{A}\mathbf{v}_{q}}{{}^{A}\boldsymbol{\omega}_{B}^{2}}. \tag{4.26}$$

Equation (4.25) can be written in the form

$$ {}^{A}\boldsymbol{\omega}_{B} \times ({}^{A}R_{B}\, {}^{B}\tilde{\mathbf{p}}) = -{}^{A}\mathbf{v}'_{q}, \tag{4.27}$$

where ${}^{A}\mathbf{v}'_{q} = {}^{A}\mathbf{v}_{q} - \lambda\, {}^{A}\boldsymbol{\omega}_{B}$ is orthogonal to ${}^{A}\boldsymbol{\omega}_{B}$; that is,

$$ {}^{A}\boldsymbol{\omega}_{B} \cdot {}^{A}\mathbf{v}'_{q} = {}^{A}\boldsymbol{\omega}_{B} \cdot ({}^{A}\mathbf{v}_{q} - \lambda\, {}^{A}\boldsymbol{\omega}_{B}) = 0. \tag{4.28}$$

We now make use of the following result derived from vector algebra. Let vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ in Fig. 4.4 satisfy the following two conditions:

$$ \mathbf{a} \times \mathbf{c} = \mathbf{b}, $$
$$ \mathbf{a} \cdot \mathbf{b} = \mathbf{0}. $$

Then $\mathbf{c}$ has infinite number of solutions lying on a line:

$$ \mathbf{c} = -\frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a}^{2}} + \mu \mathbf{a}, $$

where μ is an arbitrary scalar constant.

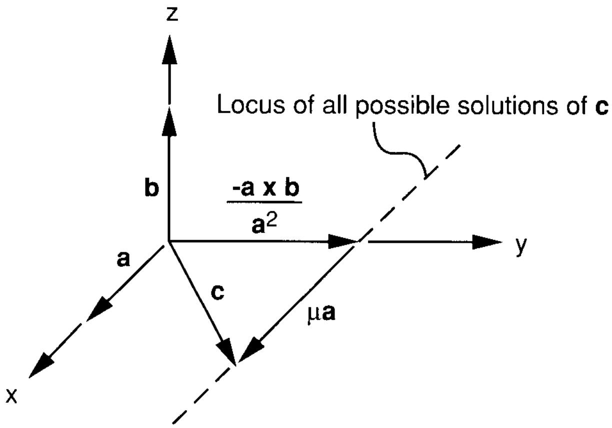

FIGURE 4.4. Vector relation.

From the vector algebra above, we conclude that all solutions to Eqs. (4.27) and (4.28) are given by

$$^{A}R_{B}\,{}^{B}\tilde{\mathbf{p}} = \frac{^{A}\boldsymbol{\omega}_{B} \times {}^{A}\mathbf{v}'_{q}}{^{A}\boldsymbol{\omega}_{B}^{2}} + \mu\,{}^{A}\boldsymbol{\omega}_{B}. \tag{4.29}$$

Applying Eq. (4.15), Eq. (4.29) can be written as

$$^{A}\tilde{\mathbf{p}} = {}^{A}\mathbf{q} + \frac{^{A}\boldsymbol{\omega}_{B} \times {}^{A}\mathbf{v}'_{q}}{^{A}\boldsymbol{\omega}_{B}^{2}} + \mu\,{}^{A}\boldsymbol{\omega}_{B}. \tag{4.30}$$

Equation (4.29) or (4.30) states that the locus of all points whose instantaneous linear velocities point along the direction of the angular velocity vector is a line. This line that is parallel to the angular velocity vector is called the *instantaneous screw axis*. We conclude that the general spatial motion of a rigid body consists of a differential rotation about, and a differential translation along, some axis.

4.3 DIFFERENTIAL KINEMATICS OF SERIAL MANIPULATORS

In this section we study the differential kinematics of a serial manipulator using the Denavit–Hartenberg transformation matrix. First, we study the differential motion of a link. Then we apply it to the differential motion of a serial manipulator.

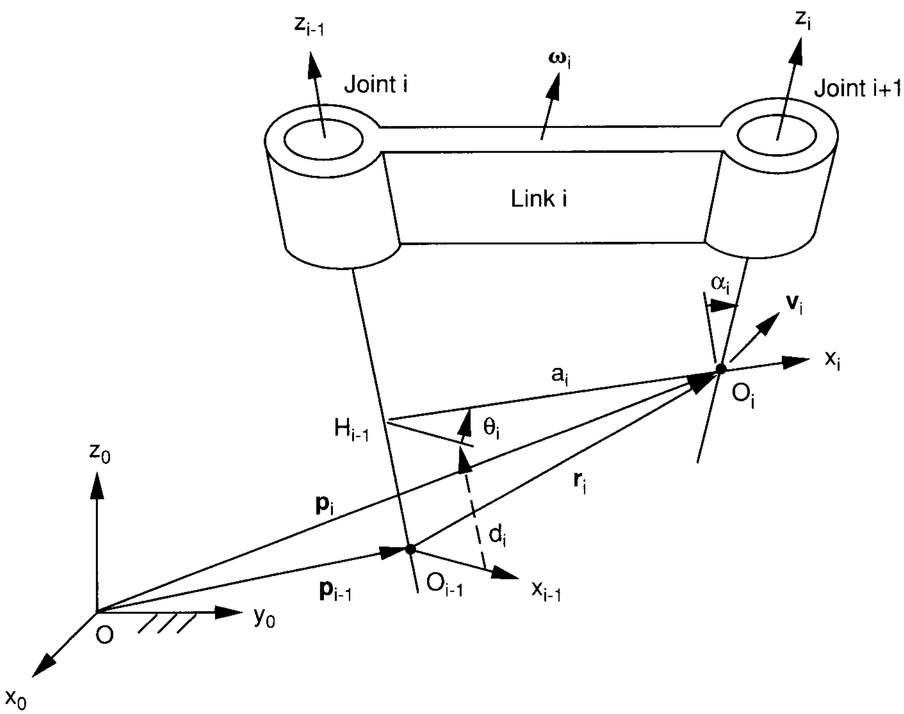

FIGURE 4.5. Geometry of link i and its motion state.

4.3.1 Link Differential Transformation Matrix

Figure 4.5 shows a typical link, i, of a manipulator. According to the D-H convention, a Cartesian coordinate system (x_i, y_i, z_i) is attached to the distal end of link i, and the fixed coordinate system is denoted by frame (x_0, y_0, z_0). The location of link i can be described by a position vector $\mathbf{p}_i$ of O_i, and a rotation matrix ${}^0R_i$ of link i with respect to the fixed reference frame 0. The velocity state of link i can be described by the linear velocity $\mathbf{v}_i$ of the origin O_i, and the angular velocity $\boldsymbol{\omega}_i$ of link i relative to the fixed reference frame.

The D-H transformation matrix is given by Eq. (2.2) and the inverse transformation is given by Eq. (2.5). Taking the derivative of Eq. (2.2) with respect to time, we obtain

$$ {}^{i-1}\dot{A}_i = \begin{bmatrix} -\dot{\theta}_i s\theta_i & -\dot{\theta}_i c\alpha_i c\theta_i & \dot{\theta}_i s\alpha_i c\theta_i & -\dot{\theta}_i a_i s\theta_i \\ \dot{\theta}_i c\theta_i & -\dot{\theta}_i c\alpha_i s\theta_i & \dot{\theta}_i s\alpha_i s\theta_i & \dot{\theta}_i a_i c\theta_i \\ 0 & 0 & 0 & \dot{d}_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4.31} $$

In Eq. (4.31), both θ_i and d_i are treated as variables. For a revolute joint, $\dot{d}_i = 0$, and for a prismatic joint, $\dot{\theta}_i = 0$. Postmultiplying both sides of Eq. (4.31) by $({}^{i-1}A_i)^{-1}$, we obtain

$$({}^{i-1}\dot{A}_i)({}^{i-1}A_i)^{-1} = \left[\begin{array}{c:c} \dot{\theta}_i\, {}^{i-1}Z_{i-1} & \dot{d}_i\, {}^{i-1}\mathbf{z}_{i-1} \\ \hdashline 0 & 0 \end{array}\right], \tag{4.32}$$

where

$${}^{i-1}\mathbf{z}_{i-1} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{4.33}$$

$${}^{i-1}Z_{i-1} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{4.34}$$

Equation (4.33) represents a unit vector pointing along the z_{i-1}-axis. Similarly, Eq. (4.34) represents a 3×3 skew-symmetric matrix whose nonzero elements denote a unit angular velocity of link i with respect to link $i-1$. Both ${}^{i-1}\mathbf{z}_{i-1}$ and ${}^{i-1}Z_{i-1}$ are expressed in the $(i-1)$th link frame. We conclude that the upper left 3×3 submatrix of $({}^{i-1}\dot{A}_i)({}^{i-1}A_i)^{-1}$ represents the angular velocity of link i relative to link $i-1$, and the fourth column represents the linear velocity of a point, which is embedded in link i and instantaneously coincident with O_{i-1}, relative to link $i-1$.

4.3.2 Overall Differential Transformation Matrix

We now apply the results derived earlier to the kinematic analysis of serial manipulators. Figure 4.6 shows a schematic of a typical serial manipulator, where $\mathbf{p}_n$ denotes the position vector of the origin of the end-effector frame, and $\mathbf{p}_{i-1}$ denotes the position vector of the origin of the $(i-1)$th frame relative to the fixed frame. Further, ${}^{i-1}\mathbf{p}_n^*$ denotes the vector pointing from O_{i-1} to O_n and expressed in the fixed frame. The loop-closure equation for such an n-dof serial manipulator can be written as

$${}^0A_n = {}^0A_1\, {}^1A_2\, {}^2A_3 \cdots {}^{n-1}A_n. \tag{4.35}$$

Taking the derivative of Eq. (4.35) with respect to time, we obtain

$$\begin{aligned} {}^0\dot{A}_n = {} & ({}^0\dot{A}_1\, {}^1A_2 \cdots {}^{n-1}A_n) + ({}^0A_1\, {}^1\dot{A}_2 \cdots {}^{n-1}A_n) \\ & + \cdots + ({}^0A_1\, {}^1A_2 \cdots {}^{n-1}\dot{A}_n). \end{aligned} \tag{4.36}$$

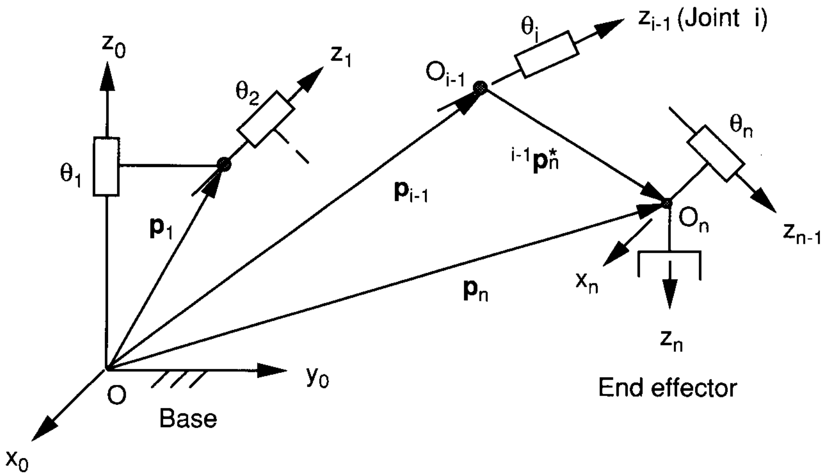

FIGURE 4.6. Link parameters of a serial manipulator.

Equation (4.36) contains 12 nontrivial scalar equations that can be reduced to a system of six independent equations as follows. Postmultiplying Eq. (4.36) by ${}^0A_n^{-1}$, we obtain

$$
\begin{aligned}
{}^0\dot{A}_n\,{}^0A_n^{-1} &= {}^0\dot{A}_1\,{}^0A_1^{-1} + {}^0A_1\,({}^1\dot{A}_2\,{}^1A_2^{-1})\,{}^0A_1^{-1} \\
&\quad + ({}^0A_1\,{}^1A_2)({}^2\dot{A}_3\,{}^2A_3^{-1})({}^0A_1\,{}^1A_2)^{-1} + \cdots.
\end{aligned} \tag{4.37}
$$

The matrix ${}^0\dot{A}_n$ can be decomposed into two submatrices:

$$
{}^0\dot{A}_n \equiv \begin{bmatrix} \dot{R}_n & \mathbf{v}_n \\ 0 & 0 \end{bmatrix}, \tag{4.38}
$$

where $\dot{R}_n$ denotes the rate of change of the end-effector rotation matrix and $\mathbf{v}_n = \dot{\mathbf{p}}_n$ denotes the linear velocity of the origin of the hand coordinate system.

Similar to Eq. (4.32), we can express the matrix products in Eq. (4.37) as

$$
{}^0\dot{A}_n\,{}^0A_n^{-1} = \begin{bmatrix} \Omega_n & \mathbf{v}_o \\ 0 & 0 \end{bmatrix}, \tag{4.39}
$$

$$
{}^{i-1}\dot{A}_i({}^{i-1}A_i)^{-1} = \begin{bmatrix} \dot{\theta}_i\,{}^{i-1}Z_{i-1} & \dot{d}_i\,{}^{i-1}\mathbf{z}_{i-1} \\ 0 & 0 \end{bmatrix}. \tag{4.40}
$$

Note that $\Omega_n = \dot{R}_n R_n^{\mathrm{T}}$ is a 3×3 skew-symmetric matrix whose elements represent the angular velocity of the end effector, and $\mathbf{v}_o$ represents the linear velocity of a point in the end effector that is instantaneously coincident with the origin of the fixed reference frame. For convenience, we define

$$^{0}A_1\,^{1}A_2\ldots{}^{i-2}A_{i-1} \equiv \begin{bmatrix} R_{i-1} & \mathbf{p}_{i-1} \\ 0 & 1 \end{bmatrix}, \tag{4.41}$$

where R_{i-1} and $\mathbf{p}_{i-1}$ denote the rotation matrix and the position vector of the origin of the $(i-1)$th frame with respect to the fixed reference frame. Substituting Eqs. (4.39) through (4.41) into (4.37) yields

$$\begin{bmatrix} \Omega_n & \mathbf{v}_o \\ 0 & 0 \end{bmatrix}$$

$$= \sum_{i=1}^{n} \left[\begin{array}{c:c} \dot{\theta}_i (R_{i-1}\,{}^{i-1}Z_{i-1} R_{i-1}^{\mathrm{T}}) & -\dot{\theta}_i (R_{i-1}\,{}^{i-1}Z_{i-1} R_{i-1}^{\mathrm{T}})\mathbf{p}_{i-1} + \dot{d}_i R_{i-1}\,{}^{i-1}\mathbf{z}_{i-1} \\ \hdashline 0 & 0 \end{array}\right]$$

$$= \sum_{i=1}^{n} \left[\begin{array}{c:c} \dot{\theta}_i Z_{i-1} & -\dot{\theta}_i Z_{i-1}\mathbf{p}_{i-1} + \dot{d}_i \mathbf{z}_{i-1} \\ \hdashline 0 & 0 \end{array}\right], \tag{4.42}$$

where

$$Z_{i-1} = R_{i-1}\,{}^{i-1}Z_{i-1}\,R_{i-1}^{\mathrm{T}}, \tag{4.43}$$

$$\mathbf{z}_{i-1} = R_{i-1}\,{}^{i-1}\mathbf{z}_{i-1}. \tag{4.44}$$

Equation (4.42) contains only six independent equations. The (3,2), (1,3), and (2,1) elements form the angular velocity vector $\boldsymbol{\omega}_n$ of the end effector, and the last column represents the linear velocity of a point in the end effector that is instantaneously coincident with the origin of the fixed frame. Writing Eq. (4.42) in vector form, we obtain

$$\boldsymbol{\omega}_n = \sum_{i=1}^{n} \dot{\theta}_i \mathbf{z}_{i-1}, \tag{4.45}$$

$$\mathbf{v}_o = \sum_{i=1}^{n} (-\dot{\theta}_i \mathbf{z}_{i-1} \times \mathbf{p}_{i-1} + \dot{d}_i \mathbf{z}_{i-1}). \tag{4.46}$$

Equations (4.45) and (4.46) imply that the angular velocities of the links are additive. We may think of the end effector as rotating instantaneously about and translating along all the joint axes, and the effect of the instantaneous motion about each joint axis can be added linearly. We note that the velocity, $\mathbf{v}_n$, of a point located at the origin of the hand coordinate system is related to $\mathbf{v}_o$ by the following transformation:

$$\mathbf{v}_n = \mathbf{v}_o + \boldsymbol{\omega}_n \times \mathbf{p}_n. \tag{4.47}$$

4.4 SCREW COORDINATES AND SCREW SYSTEMS

We have shown that both finite and infinitesimal displacements of a rigid body can conveniently be expressed as a rotation about a unique axis and a translation along the same axis. This combined motion is called a *screw displacement* or *twist*, and the unique axis is called a *screw axis* of the displacement. The ratio of translation to rotation is called the *pitch*, λ. Specifically, $\lambda = d/\theta$ for finite displacements and $\lambda = \dot{d}/\dot{\theta}$ for infinitesimal displacements.

A screw can conveniently be represented by a coordinate system called the *screw coordinates*. We define the coordinates of a unit screw, $\hat{\$}$, by a pair of vectors (Ball, 1900; Beyer, 1958; Dimentberg, 1965; Roth, 1984):

$$\hat{\$} = \begin{bmatrix} \mathbf{s} \\ \mathbf{s}_o \times \mathbf{s} + \lambda \mathbf{s} \end{bmatrix} = \begin{bmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \\ S_5 \\ S_6 \end{bmatrix}, \tag{4.48}$$

where $\mathbf{s}$ is a unit vector pointing along the direction of the screw axis and $\mathbf{s}_o$ is the position vector of any point on the screw axis.

The vector $\mathbf{s}_o \times \mathbf{s}$ defines the moment of the screw axis about the origin of a reference frame. Since the screw axis and its moment are directed at right angles to one another, $\mathbf{s} \cdot (\mathbf{s}_o \times \mathbf{s}) = 0$ identically. Hence only five of the six coordinates are independent.

For a revolute joint, $\lambda = 0$, the unit screw reduces to

$$\hat{\$} = \begin{bmatrix} \mathbf{s} \\ \mathbf{s}_o \times \mathbf{s} \end{bmatrix}. \tag{4.49}$$

For a prismatic joint, $\lambda = \infty$, the unit screw is defined as

$$\hat{\$} = \begin{bmatrix} \mathbf{0} \\ \mathbf{s} \end{bmatrix}. \tag{4.50}$$

The screw axis and the pitch together determine the screw. However, the displacement is not determined completely until after the *amplitude* or *inten-*

sity of the screw axis is specified. Let $\dot{q}$ be the intensity of a twist. Then the twist can be written as

$$\$ = \dot{q}\hat{\$}, \tag{4.51}$$

where $\dot{q} = \dot{\theta}$ for a revolute joint and $\dot{q} = \dot{d}$ for a prismatic joint.

Hence six screw coordinates completely specify the first-order instantaneous kinematics of a rigid body. The first three coordinates of $\$$ represent the angular velocity, and the last three coordinates represent the linear velocity of a point that is embedded in the moving body and instantaneously coincident with the origin of the fixed reference frame.

We may consider the motion of a rigid body as being twisted instantaneously about several screw axes. These screws form a screw system in three-dimensional space (Hunt, 1978). The screw system that permits a rigid body to move instantaneously with 1 degree of freedom consists of a single screw. This is called a *first-order screw system* or simply a *1-system*.

The infinitesimal motion of a rigid body with 2 degrees of freedom can generally be considered as the resulting motion of two instantaneous screws of arbitrary pitches. In fact, we can visualize the body as being connected to a fixed base by two joints with one intermediate link. These two joint axes can be arbitrarily located in space. Given the intensities of the two screws, the instantaneous velocity vectors of all points in the body are completely known, and this velocity state can be represented by a resulting twist of prescribed pitch and intensity. It can be shown that for any instantaneous motion, the ratio of the two intensities determine the location and pitch of the resulting twist, while the actual intensities determine the intensity of the resulting twist. As the intensities of these two screws change, a single infinitude of resulting twists will be generated. The axes of these screws form a line series in space known as the *screw cylindroid*. We call this system of infinite many screws a *2-system*. Similarly, the infinitesimal motion of a rigid body with n degrees of freedom, $n \leq 6$, can generally be described by an n-system.

For a serial manipulator, we may consider the motion of the end effector as being twisted instantaneously about the joint axes of an open-loop chain. These instantaneous twists may be added linearly to give the resulting motion of the end effector. Thus the first-order instantaneous kinematics can be written as

$$\$_n = \sum_{i=1}^{n} \dot{q}_i \hat{\$}_i, \tag{4.52}$$

where $\hat{\$}_i$ is a unit screw defined by the joints of the manipulator.

We note that a line in three-dimensional space can be represented by a coordinate system called the *Plücker* or *line* coordinates. If the Plücker coordinates of a line are $[L, M, N, P, Q, R]^{\mathrm{T}}$, the Plücker coordinates are related to the screw coordinates by

$$\begin{bmatrix} L \\ M \\ N \\ P \\ Q \\ R \end{bmatrix} = \begin{bmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 - \lambda S_1 \\ S_5 - \lambda S_2 \\ S_6 - \lambda S_3 \end{bmatrix}. \tag{4.53}$$

4.5 MANIPULATOR JACOBIAN MATRIX

Let $x_i = f_i(q_1, q_2, q_3, \ldots, q_n)$ for $i = 1, 2, 3, \ldots, m$ be a set of m equations, each a function of n independent variables. Then the time derivatives of x_i can be written as a function of $\dot{q}_i$ as follows:

$$\dot{x}_i = \frac{\partial f_i}{\partial q_1}\dot{q}_1 + \frac{\partial f_i}{\partial q_2}\dot{q}_2 + \frac{\partial f_i}{\partial q_3}\dot{q}_3 + \cdots + \frac{\partial f_i}{\partial q_n}\dot{q}_n, \qquad i = 1, 2, 3, \ldots, m. \tag{4.54}$$

Writing Eq. (4.54) in matrix form, we obtain

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_m \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f_1}{\partial q_1} & \dfrac{\partial f_1}{\partial q_2} & \cdots & \dfrac{\partial f_1}{\partial q_n} \\ \dfrac{\partial f_2}{\partial q_1} & \dfrac{\partial f_2}{\partial q_2} & \cdots & \dfrac{\partial f_2}{\partial q_n} \\ \vdots & \vdots & \cdots & \vdots \\ \dfrac{\partial f_m}{\partial q_1} & \dfrac{\partial f_m}{\partial q_2} & \cdots & \dfrac{\partial f_m}{\partial q_n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}, \tag{4.55}$$

or simply

$$\dot{\mathbf{x}} = J\dot{\mathbf{q}}, \tag{4.56}$$

where $\mathbf{x} = [x_1, x_2, \ldots, x_m]^{\mathrm{T}}$ denotes an m-dimensional vector, $\mathbf{q} = [q_1, q_2, \ldots, q_n]^{\mathrm{T}}$ denotes an n-dimensional vector, and J denotes the $m \times n$ matrix of the partial derivatives in Eq. (4.55).

We call J the *Jacobian matrix*, or simply *Jacobian*. The Jacobian matrix is a linear transformation matrix that maps an n-dimensional velocity vector $\dot{\mathbf{q}}$ into an m-dimensional velocity vector $\dot{\mathbf{x}}$. We may think of the elements of J as the influence coefficients of the vector function $\mathbf{x}$. The (i, j) element of J

describes how a differential change in q_j affects the differential change in x_i. In general, the vector $\mathbf{x}$ is a nonlinear function of $\mathbf{q}$. Hence the Jacobian matrix is also a function of $\mathbf{q}$. Thus, the Jacobian matrix is configuration dependent.

For robot manipulators, the Jacobian matrix is defined as the coefficient matrix of any set of equations that relates the velocity state of the end effector to the actuated joint rates. The joint rates are defined as

$$\dot{q}_i = \begin{cases} \dot{\theta}_i & \text{for a revolute joint.} \\ \dot{d}_i & \text{for a prismatic joint.} \end{cases} \tag{4.57}$$

The velocity state of the end effector, $\mathbf{x}$, can be expressed in several different ways. Perhaps the most commonly used definitions are the *conventional Jacobian* and *screw-based Jacobian.*

1. *Conventional Jacobian.* In a conventional Jacobian, the end-effector velocity state is expressed in terms of the linear velocity of the origin of the end-effector coordinate frame, $\mathbf{v}_n$, and the angular velocity of the end effector, $\boldsymbol{\omega}_n$.

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_n \\ \boldsymbol{\omega}_n \end{bmatrix}. \tag{4.58}$$

2. *Screw-based Jacobian.* The screw-based Jacobian is defined in terms of the angular velocity of the end effector, $\boldsymbol{\omega}_n$, and the linear velocity of a reference point, $\mathbf{v}_o$, in the end effector that is instantaneously coincident with the origin of a reference frame in which the screws are expressed:

$$\dot{\mathbf{x}} = \begin{bmatrix} \boldsymbol{\omega}_n \\ \mathbf{v}_o \end{bmatrix}. \tag{4.59}$$

We note that the end-effector velocity, $\dot{\mathbf{x}}$, for the screw-based Jacobian is defined with its angular and linear velocity vectors arranged in reverse order from the conventional Jacobian.

In general, the Jacobian matrix is an $m \times n$ matrix, where m denotes the degrees of freedom of the end-effector space and n denotes the number of actuated joint variables. For a 6-dof spatial manipulator, $m = n = 6$, the Jacobian matrix is a 6×6 square matrix. For a manipulator with less than 6 degrees of freedom, the end-effector velocity state may contain just the linear velocity vector, or the angular velocity vector, or a combination of some linear and angular velocity components. For example, the working space of a planar manipulator is confined to a two-dimensional space. A three-component vector $\dot{\mathbf{x}} = [v_x, v_y, \omega_z]^{\mathrm{T}}$ is sufficient to describe the velocity state of the end effector. Hence the Jacobian reduces to a 3×3 matrix. Similarly, for a point-positioning device $\dot{\mathbf{x}} = [v_x, v_y, v_z]^T$, and for a body-orienting mechanism $\dot{\mathbf{x}} = [\omega_x, \omega_y, \omega_z]^T$. On the other hand, for a manipulator with redundant degrees of freedom, we may have $n > 6$. In this book, we concentrate on nonredundant robot manipulators.

4.6 CONVENTIONAL JACOBIAN

As mentioned earlier, any point in the end effector can be chosen as the reference point to describe the velocity state of the end effector. A logical choice is the origin, O_n, of the end-effector frame (Whitney, 1972). Using this definition, the end-effector velocity state can be expressed in terms of the joint rates as follows:

$$\mathbf{v}_n = \sum_{i=1}^{n} [\dot{\theta}_i (\mathbf{z}_{i-1} \times {}^{i-1}\mathbf{p}_n^*) + \mathbf{z}_{i-1}\dot{d}_i] \tag{4.60}$$

$$\boldsymbol{\omega}_n = \sum_{i=1}^{n} \dot{\theta}_i \mathbf{z}_{i-1} \tag{4.61}$$

where $\dot{\theta}_i$ and $\dot{d}_i$ are the rate of rotation about and translation along the ith joint axis, $\mathbf{z}_{i-1}$ is a unit vector along the ith joint axis, and ${}^{i-1}\mathbf{p}_n^*$ is a vector defined from the origin of the $(i-1)$th link frame, O_{i-1}, to the origin of the end effector frame, as shown in Fig. 4.6. Note that all vectors in Eqs. (4.60) and (4.61) are expressed in the fixed coordinate frame, (x_0, y_0, z_0).

Writing Eqs. (4.60) and (4.61) in matrix form, we obtain

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{v}_n \\ \boldsymbol{\omega}_n \end{bmatrix} = J\dot{\mathbf{q}}, \tag{4.62}$$

where

$$J = [J_1, J_2, \ldots, J_n],$$

$$J_i = \begin{bmatrix} \mathbf{z}_{i-1} \times {}^{i-1}\mathbf{p}_n^* \\ \mathbf{z}_{i-1} \end{bmatrix} \quad \text{for a revolute joint,}$$

$$J_i = \begin{bmatrix} \mathbf{z}_{i-1} \\ \mathbf{0} \end{bmatrix} \quad \text{for a prismatic joint.}$$

The left-hand side of Eq. (4.62) is a 6×1 vector composed of the elements of $\mathbf{v}_n$ and $\boldsymbol{\omega}_n$, while the right-hand side is a product of the Jacobian matrix and the vector of joint rates. The vector of joint rates consists of all the actuated joint rates, $\dot{q}_i$ for $i = 1, 2, \ldots, n$. The ith column of the Jacobian matrix,

J_i, represents the effect of the ith joint rate on the velocity state of the end effector.

Equation (4.62) implies that to compute the Jacobian matrix, the direction and location of each joint axis should be determined first. This can be accomplished by the following matrix operations:

$$\mathbf{z}_{i-1} = {}^0R_{i-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{4.63}$$

$$^{i-1}\mathbf{p}_n^* = {}^0R_{i-1}\, {}^{i-1}\mathbf{r}_i + {}^i\mathbf{p}_n^*, \tag{4.64}$$

where

$$^{i-1}\mathbf{r}_i = \begin{bmatrix} a_i c\theta_i \\ a_i s\theta_i \\ d_i \end{bmatrix}$$

denotes the vector $\overline{O_{i-1}O_i}$ expressed in the $(i-1)$th link frame, while $^{i-1}\mathbf{p}_n^*$ denotes the vector $\overline{O_{i-1}O_n}$ expressed in the fixed frame. Once the Jacobian is known, the end-effector velocity can be computed directly from Eq. (4.62) for any given joint rates. On the other hand, given a desired end-effector velocity, the inverse transformation of Eq. (4.62) can be solved for the joint rates.

4.6.1 Jacobian of a Planar 2-DOF Manipulator

In this example we derive the Jacobian of a planar 2-dof manipulator shown in Fig. 4.7. The manipulator is made up of two revolute joints, with both axes pointing out of the paper. A coordinate system is attached to each link according to the D-H convention for the purpose of analysis. The (x_0, y_0) coordinate system is attached to the base with its origin located at the fixed pivot O. The x_0-axis points to the right.

We first compute the vectors $\mathbf{z}_{i-1}$ and $^{i-1}\mathbf{p}_2^*$ by applying Eqs. (4.63) and (4.64):

$$\mathbf{z}_0 = \mathbf{z}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$^1\mathbf{p}_2^* = \begin{bmatrix} a_2 c\theta_{12} \\ a_2 s\theta_{12} \\ 0 \end{bmatrix},$$

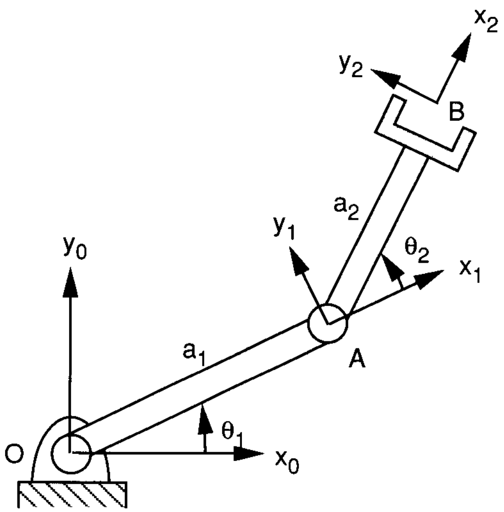

FIGURE 4.7. Planar 2-dof, $2R$ manipulator.

$$
{}^{0}\mathbf{p}_2^* = \begin{bmatrix} a_1 c\theta_1 + a_2 c\theta_{12} \\ a_1 s\theta_1 + a_2 s\theta_{12} \\ 0 \end{bmatrix},
$$

where $\theta_{12} = \theta_1 + \theta_2$. We note that the expressions above can be obtained directly from the geometry of the links without using the Denavit–Hartenberg transformation matrices. Substituting the expressions above into Eq. (4.62), we obtain

$$
\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} -a_1 s\theta_1 - a_2 s\theta_{12} & -a_2 s\theta_{12} \\ a_1 c\theta_1 + a_2 c\theta_{12} & a_2 c\theta_{12} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}. \tag{4.65}
$$

Hence the Jacobian matrix is given by

$$
J = \begin{bmatrix} -a_1 s\theta_1 - a_2 s\theta_{12} & -a_2 s\theta_{12} \\ a_1 c\theta_1 + a_2 c\theta_{12} & a_2 c\theta_{12} \end{bmatrix}. \tag{4.66}
$$

4.6.2 Jacobian of a Planar 3-DOF Manipulator

As a second example, we study the conventional Jacobian of the planar 3-dof manipulator shown in Fig. 2.3. We first compute the vectors $\mathbf{z}_{i-1}$ and ${}^{i-1}\mathbf{p}_3^*$ from Eqs. (4.63) and (4.64), for $i = 1, 2$, and 3 as follows:

$$\mathbf{z}_0 = \mathbf{z}_1 = \mathbf{z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

$$^2\mathbf{p}_3^* = \begin{bmatrix} a_3 c\theta_{123} \\ a_3 s\theta_{123} \\ 0 \end{bmatrix},$$

$$^1\mathbf{p}_3^* = \begin{bmatrix} a_2 c\theta_{12} + a_3 c\theta_{123} \\ a_2 s\theta_{12} + a_3 s\theta_{123} \\ 0 \end{bmatrix},$$

$$^0\mathbf{p}_3^* = \begin{bmatrix} a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123} \\ a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123} \\ 0 \end{bmatrix},$$

where $\theta_{12} = \theta_1 + \theta_2$ and $\theta_{123} = \theta_1 + \theta_2 + \theta_3$. Substituting the expressions above into Eq. (4.62), we obtain

$$\begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} = J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix},$$

where

$$J = \begin{bmatrix} -(a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123}) & -(a_2 s\theta_{12} + a_3 s\theta_{123}) & -a_3 s\theta_{123} \\ (a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123}) & (a_2 c\theta_{12} + a_3 c\theta_{123}) & a_3 c\theta_{123} \\ 1 & 1 & 1 \end{bmatrix}. \tag{4.67}$$

We note that if the reference point is chosen at origin of the (x_2, y_2) frame, the Jacobian matrix reduces to

$$J = \begin{bmatrix} -(a_1 s\theta_1 + a_2 s\theta_{12}) & -(a_2 s\theta_{12}) & 0 \\ (a_1 c\theta_1 + a_2 c\theta_{12}) & (a_2 c\theta_{12}) & 0 \\ 1 & 1 & 1 \end{bmatrix}. \tag{4.68}$$

4.6.3 Jacobian of the Stanford Manipulator

Figure 4.8 shows a schematic diagram of the Stanford arm described in Chapter 2. To simplify the analysis, the origin of the fixed coordinate frame is located at the point of intersection of the first two joint axes, and the origin of the (x_6, y_6, z_6) frame is located at the point of intersection of the last three joint axes.

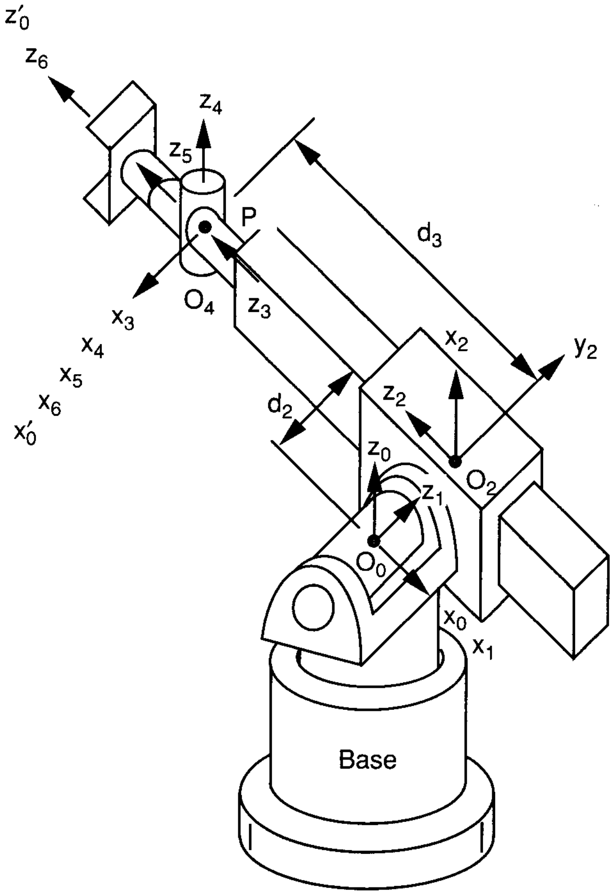

FIGURE 4.8. Stanford manipulator.

The D-H link parameters are listed in Table 4.1, from which the D-H transformation matrices are derived as follows:

$$
{}^0A_1 = \begin{bmatrix} c\theta_1 & 0 & -s\theta_1 & 0 \\ s\theta_1 & 0 & c\theta_1 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^1A_2 = \begin{bmatrix} c\theta_2 & 0 & s\theta_2 & 0 \\ s\theta_2 & 0 & -c\theta_2 & 0 \\ 0 & 1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^2A_3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^3A_4 = \begin{bmatrix} c\theta_4 & 0 & -s\theta_4 & 0 \\ s\theta_4 & 0 & c\theta_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

TABLE 4.1. D-H Link Parameters of the Stanford Arm

Joint i	α_i	a_i	d_i	θ_i
1	$-90°$	0	0	θ_1 (variable)
2	$90°$	0	d_2 (constant)	θ_2 (variable)
3	$0°$	0	d_3 (variable)	$-90°$ (constant)
4	$-90°$	0	0	θ_4 (variable)
5	$90°$	0	0	θ_5 (variable)
6	$0°$	0	0	θ_6 (variable)

$$
{}^4A_5 = \begin{bmatrix} c\theta_5 & 0 & s\theta_5 & 0 \\ s\theta_5 & 0 & -c\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^5A_6 = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ s\theta_6 & c\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

The directions of the joint axes, $\mathbf{z}_{i-1}$, are derived by applying Eq. (4.63):

$$
\mathbf{z}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{4.69}
$$

$$
\mathbf{z}_1 = {}^0R_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -s\theta_1 \\ c\theta_1 \\ 0 \end{bmatrix}, \tag{4.70}
$$

$$
\mathbf{z}_2 = \mathbf{z}_3 = {}^0R_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c\theta_1 s\theta_2 \\ s\theta_1 s\theta_2 \\ c\theta_2 \end{bmatrix}, \tag{4.71}
$$

$$
\mathbf{z}_4 = {}^0R_4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -s\theta_1 s\theta_4 + c\theta_1 c\theta_2 c\theta_4 \\ c\theta_1 s\theta_4 + s\theta_1 c\theta_2 c\theta_4 \\ -s\theta_2 c\theta_4 \end{bmatrix}, \tag{4.72}
$$

$$
\mathbf{z}_5 = {}^0R_5 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} s\theta_1 c\theta_4 s\theta_5 + c\theta_1 c\theta_2 s\theta_4 s\theta_5 + c\theta_1 s\theta_2 c\theta_5 \\ -c\theta_1 c\theta_4 s\theta_5 + s\theta_1 c\theta_2 s\theta_4 s\theta_5 + s\theta_1 s\theta_2 c\theta_5 \\ -s\theta_2 s\theta_4 s\theta_5 + c\theta_2 c\theta_5 \end{bmatrix}. \tag{4.73}
$$

The position vectors, ${}^{i-1}\mathbf{p}_6^*$, are derived by applying Eq. (4.64):

$$
{}^3\mathbf{p}_6^* = {}^4\mathbf{p}_6^* = {}^5\mathbf{p}_6^* = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \tag{4.74}
$$

$$
{}^{2}\mathbf{p}_6^* = \begin{bmatrix} d_3 c\theta_1 s\theta_2 \\ d_3 s\theta_1 s\theta_2 \\ d_3 c\theta_2 \end{bmatrix}, \tag{4.75}
$$

$$
{}^{1}\mathbf{p}_6^* = \begin{bmatrix} d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 \\ d_3 s\theta_1 s\theta_2 + d_2 c\theta_1 \\ d_3 c\theta_2 \end{bmatrix}, \tag{4.76}
$$

$$
{}^{0}\mathbf{p}_6^* = \begin{bmatrix} d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 \\ d_3 s\theta_1 s\theta_2 + d_2 c\theta_1 \\ d_3 c\theta_2 \end{bmatrix}. \tag{4.77}
$$

The Jacobian matrix is derived by applying Eq. (4.62) column by column:

$$
\begin{bmatrix} v_{6x} \\ v_{6y} \\ v_{6z} \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = J \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \\ \dot{\theta}_6 \end{bmatrix},
$$

where the Jacobian matrix is given by

$$
J = \begin{bmatrix} -d_3 s\theta_1 s\theta_2 - d_2 c\theta_1 & d_3 c\theta_1 c\theta_2 & c\theta_1 s\theta_2 & 0 & 0 & 0 \\ d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 & d_3 s\theta_1 c\theta_2 & s\theta_1 s\theta_2 & 0 & 0 & 0 \\ 0 & -d_3 s\theta_2 & c\theta_2 & 0 & 0 & 0 \\ 0 & -s\theta_1 & 0 & c\theta_1 s\theta_2 & j_{45} & j_{46} \\ 0 & c\theta_1 & 0 & s\theta_1 s\theta_2 & j_{55} & j_{56} \\ 1 & 0 & 0 & c\theta_2 & j_{65} & j_{66} \end{bmatrix}, \tag{4.78}
$$

where j_{45}, j_{55}, and j_{65} represent the x, y, and z components of the unit vector $\mathbf{z}_4$, and j_{46}, j_{56}, and j_{66} are the x, y, and z components of $\mathbf{z}_5$.

As a consequence of the concurrence of axes 4, 5 and 6, all the elements in the upper right 3×3 submatrix are equal to zero.

4.7 SCREW-BASED JACOBIAN

In the preceding section, the origin of the hand coordinate system was chosen as the reference point. It turns out that it is more convenient to use the origin of the fixed frame as the reference point. In this way, the screw coordinates can be used for formulating the Jacobian matrix (Hunt, 1986, 1987a; Sugimoto, 1984; Waldron, 1982). Let $\mathbf{v}_o$ be the velocity of a point in the end-effector of a serial manipulator that is instantaneously coincident with the origin of the fixed reference frame. Then we can apply Eq. (4.52) to get the end-effector velocity vector as

$$\dot{\mathbf{x}} = \begin{bmatrix} \omega_n \\ \mathbf{v}_o \end{bmatrix} = \sum_{i=1}^{n} \dot{q}_i \hat{\$}_i, \tag{4.79}$$

where the unit twist is defined by Eq. (4.49) or (4.50).

We observe from Eq. (4.79) that the Jacobian matrix is simply an assemblage of the unit screws associated with the joint axes of a manipulator. Namely, the coordinates of the unit joint screws are the columns of the Jacobian matrix:

$$J = \begin{bmatrix} \hat{\$}_1, \hat{\$}_2, \ldots, \hat{\$}_n \end{bmatrix}. \tag{4.80}$$

Therefore, to compute the Jacobian matrix, the directions and the locations of the joint axes relative to a reference frame should be determined first. This can often be accomplished by an inspection of the geometry of the manipulator. On the other hand, if the link coordinate systems, (x_0, y_0, z_0) to (x_6, y_6, z_6), are established in accordance with the Denavit–Hartenberg convention, $\mathbf{s}_i$ and $\mathbf{s}_{oi}$ can be taken from the third and fourth columns of ${}^0A_{i-1}$.

Note that the coordinates of the joint screws can be described in any reference frame. It turns out that the Jacobian matrix can be greatly simplified if a reference frame is defined at a location where it is instantaneously coincident with one of the intermediate link frames, typically link 3 or 4 (Hunt, 1987b; Waldron et al., 1985). Specifically, if the link geometry contains concurrent joint axes, it is advantageous to locate the origin of the instantaneous reference frame at the point of concurrence. Similarly, if the link geometry contains parallel joint axes, it is preferable to align the coordinate axes of the instantaneous reference frame with these parallel axes and their common perpendiculars.

Let an instantaneous reference frame, (x'_0, y'_0, z'_0), be coincident with the (x_j, y_j, z_j) link frame as shown in Fig. 4.9. Then the directions and locations of the joint axes relative to the (x'_0, y'_0, z'_0) reference frame are given by the third and fourth column of ${}^jA_i$, respectively. That is,

$$\begin{bmatrix} \mathbf{s}_{i+1} \\ 0 \end{bmatrix} = {}^jA_i \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \tag{4.81}$$

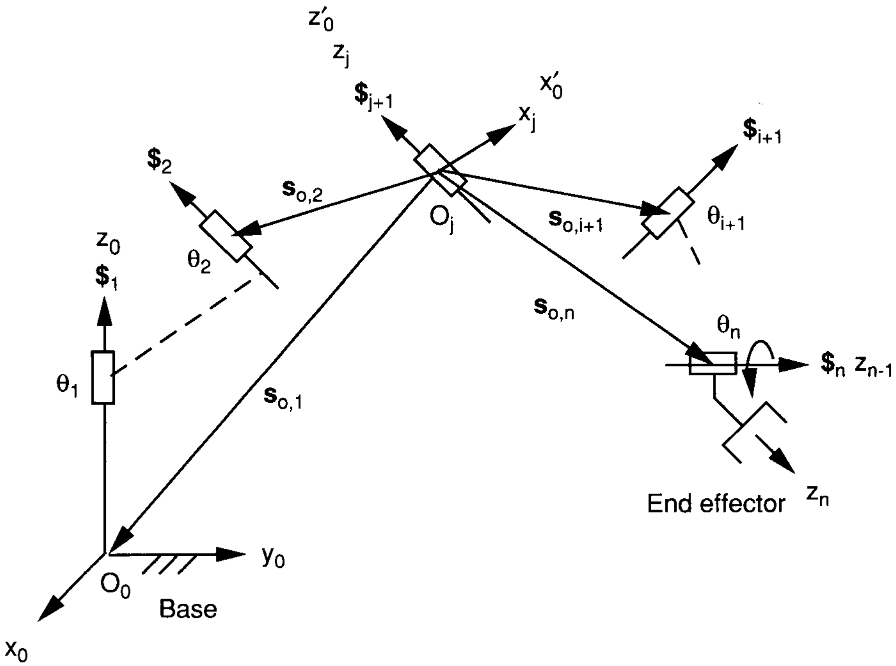

FIGURE 4.9. Screw coordinates with respect to an instantaneous reference frame.

$$\begin{bmatrix} \mathbf{s}_{o,i+1} \\ 1 \end{bmatrix} = {}^{j}A_i \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \tag{4.82}$$

The screw coordinates above can be computed efficiently by using the following recursive formulas:

1. *Initial conditions.* We begin with $\mathbf{s}_{j+1} = [0, 0, 1]^{\mathrm{T}}$ and $\mathbf{s}_{o,j+1} = [0, 0, 0]^{\mathrm{T}}$.
2. *Forward computation.* For $i = j + 1, j + 2$, to $n - 1$, we compute

$$\begin{aligned} \mathbf{s}_{i+1} &= ({}^{j}R_i)({}^{i}\mathbf{z}_i), \\ \mathbf{s}_{o,i+1} &= \mathbf{s}_{o,i} + ({}^{j}R_i)({}^{i}\mathbf{r}_i), \\ {}^{j}R_{i+1} &= ({}^{j}R_i)({}^{i}R_{i+1}). \end{aligned} \tag{4.83}$$

3. *Backward computation.* For $i = j - 1, j - 2$, to 0, we compute

$$\begin{aligned}
\mathbf{s}_{i+1} &= ({}^jR_i)({}^i\mathbf{z}_i), \\
\mathbf{s}_{o,i+1} &= \mathbf{s}_{o,i+2} - ({}^jR_{i+1})({}^{i+1}\mathbf{r}_{i+1}), \\
{}^jR_{i-1} &= ({}^jR_i)({}^iR_{i-1}),
\end{aligned} \tag{4.84}$$

where ${}^{i-1}R_i$ is the rotation matrix of link i with respect to link $i-1$, ${}^iR_{i-1} = ({}^{i-1}R_i)^{\mathrm{T}}$,

$${}^i\mathbf{z}_i = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

is a unit vector pointing along the z_i-axis and expressed in the ith link frame, and

$${}^i\mathbf{r}_i = \begin{bmatrix} a_i \\ d_i s\alpha_i \\ d_i c\alpha_i \end{bmatrix}$$

is the position vector defined from O_{i-1} to O_i and expressed in the ith link frame.

Assembling the unit screws derived from the recursive formulas above, we obtain

$$\begin{bmatrix} {}^j\boldsymbol{\omega}_n \\ {}^j\mathbf{v}_o \end{bmatrix} = {}^jJ\dot{\mathbf{q}}. \tag{4.85}$$

The leading superscript, j, indicates that the velocity state of the end effector is expressed in a reference frame that is instantaneously coincident with the jth link frame. Given the joint rates, Eq. (4.85) can be used to compute for the velocity state of the end effector. On the other hand, given a desired velocity state of the end effector, one can solve the inverse transformation of Eq. (4.85) for the joint rate. This can best be illustrated by the following examples.

4.7.1 Screw-based Jacobian of the Stanford Manipulator

To illustrate the methodology, we derive the screw-based Jacobian of the Stanford manipulator. Let the instantaneous reference frame, (x'_0, y'_0, z'_0), be coincident with the (x_3, y_3, z_3) link frame as shown in Fig. 4.8. The initial conditions for $j = 3$ are $\mathbf{s}_4 = [0, 0, 1]^{\mathrm{T}}$ and $\mathbf{s}_{o,4} = [0, 0, 0]^{\mathrm{T}}$. Applying Eq. (4.83), we obtain the directions and locations of the fifth and sixth joint axes as follows. For $i = 4$, we obtain

$$\mathbf{s}_5 = {}^3R_4\,{}^4\mathbf{z}_4 = \begin{bmatrix} -s\theta_4 \\ c\theta_4 \\ 0 \end{bmatrix}, \tag{4.86}$$

$$\mathbf{s}_{o,5} = \mathbf{s}_{o,4} + {}^3R_4\,{}^4r_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \tag{4.87}$$

$${}^3R_5 = {}^3R_4\,{}^4R_5 = \begin{bmatrix} c\theta_4 c\theta_5 & -s\theta_4 & c\theta_4 s\theta_5 \\ s\theta_4 c\theta_5 & c\theta_4 & s\theta_4 s\theta_5 \\ -s\theta_5 & 0 & c\theta_5 \end{bmatrix}. \tag{4.88}$$

For $i = 5$, we obtain

$$\mathbf{s}_6 = {}^3R_5\,{}^5\mathbf{z}_5 = \begin{bmatrix} c\theta_4 s\theta_5 \\ s\theta_4 s\theta_5 \\ c\theta_5 \end{bmatrix}, \tag{4.89}$$

$$\mathbf{s}_{o,6} = \mathbf{s}_{o,5} + {}^3R_5\,{}^5r_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \tag{4.90}$$

Applying Eq. (4.84), we obtain the directions and locations of the third, second, and first joint axes in turn as follows. For $i = 2$, we obtain

$$\mathbf{s}_3 = {}^3R_2\,{}^2\mathbf{z}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{4.91}$$

$$\mathbf{s}_{o,3} = \mathbf{s}_{o,4} - {}^3R_3\,{}^3r_3 = \begin{bmatrix} 0 \\ 0 \\ -d_3 \end{bmatrix}, \tag{4.92}$$

$${}^3R_1 = {}^3R_2\,{}^2R_1 = \begin{bmatrix} 0 & 0 & -1 \\ c\theta_2 & s\theta_2 & 0 \\ s\theta_2 & -c\theta_2 & 0 \end{bmatrix}. \tag{4.93}$$

For $i = 1$, we obtain

$$\mathbf{s}_2 = {}^3R_1\,{}^1\mathbf{z}_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \tag{4.94}$$

$$\mathbf{s}_{o,2} = \mathbf{s}_{o,3} - {}^3R_2\,{}^2r_2 = \begin{bmatrix} d_2 \\ 0 \\ -d_3 \end{bmatrix}, \tag{4.95}$$

$$^3R_0 = {^3R_1}\,{^1R_0} = \begin{bmatrix} s\theta_1 & -c\theta_1 & 0 \\ c\theta_1 c\theta_2 & s\theta_1 c\theta_2 & -s\theta_2 \\ c\theta_1 s\theta_2 & s\theta_1 s\theta_2 & c\theta_2 \end{bmatrix}. \tag{4.96}$$

For $i = 0$, we obtain

$$\mathbf{s}_1 = {^3R_0}\,{^0\mathbf{z}_0} = \begin{bmatrix} 0 \\ -s\theta_2 \\ c\theta_2 \end{bmatrix}, \tag{4.97}$$

$$\mathbf{s}_{o,1} = \mathbf{s}_{o,2} - {^3R_1}\,{^1r_1} = \begin{bmatrix} d_2 \\ 0 \\ -d_3 \end{bmatrix}. \tag{4.98}$$

We now apply Eqs. (4.49) and (4.50), column by column, to compute the Jacobian matrix. The result is

$$^3J = \begin{bmatrix} 0 & -1 & 0 & 0 & -s\theta_4 & c\theta_4 s\theta_5 \\ -s\theta_2 & 0 & 0 & 0 & c\theta_4 & s\theta_4 s\theta_5 \\ c\theta_2 & 0 & 0 & 1 & 0 & c\theta_5 \\ -d_3 s\theta_2 & 0 & 0 & 0 & 0 & 0 \\ -d_2 c\theta_2 & d_3 & 0 & 0 & 0 & 0 \\ -d_2 s\theta_2 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}. \tag{4.99}$$

We observe that the Jacobian matrix is greatly simplified by expressing it in the instantaneous frame of reference. Substituting Eq. (4.99) into Eq. (4.85), we obtain

$$\begin{aligned} {^3\omega_x} &= -\dot\theta_2 - \dot\theta_5 s\theta_4 + \dot\theta_6 c\theta_4 s\theta_5, \\ {^3\omega_y} &= -\dot\theta_1 s\theta_2 + \dot\theta_5 c\theta_4 + \dot\theta_6 s\theta_4 s\theta_5, \\ {^3\omega_z} &= \dot\theta_1 c\theta_2 + \dot\theta_4 + \dot\theta_6 c\theta_5, \\ {^3v_{ox}} &= -\dot\theta_1 d_3 s\theta_2, \\ {^3v_{oy}} &= -\dot\theta_1 d_2 c\theta_2 + \dot\theta_2 d_3, \\ {^3v_{oz}} &= -\dot\theta_1 d_2 s\theta_2 + \dot d_3. \end{aligned} \tag{4.100}$$

The inverse transformation of Eq. (4.100) can be easily derived as

$$\dot\theta_1 = -\frac{^3v_{ox}}{d_3 s\theta_2},$$

$$\dot\theta_2 = \frac{^3v_{oy} + \dot\theta_1 d_2 c\theta_2}{d_3},$$

$$\begin{aligned}
\dot{d}_3 &= {}^3v_{oz} + \dot{\theta}_1 d_2 s\theta_2, \\
\dot{\theta}_5 &= -{}^3\omega_x s\theta_4 + {}^3\omega_y c\theta_4 + \dot{\theta}_1 s\theta_2 c\theta_4 - \dot{\theta}_2 s\theta_4, \\
\dot{\theta}_6 &= \frac{{}^3\omega_x c\theta_4 + {}^3\omega_y s\theta_4 + \dot{\theta}_1 s\theta_2 s\theta_4 + \dot{\theta}_2 c\theta_4)}{s\theta_5}, \\
\dot{\theta}_4 &= {}^3\omega_z - \dot{\theta}_1 c\theta_2 - \dot{\theta}_6 c\theta_5.
\end{aligned} \tag{4.101}$$

Hence given the joint rates $\dot{\mathbf{q}} = [\dot{\theta}_1, \dot{\theta}_2, \dot{d}_3, \dot{\theta}_4, \dot{\theta}_5, \dot{\theta}_6]$, the end-effector velocity state, ${}^3\boldsymbol{\omega}_n$ and ${}^3\mathbf{v}_o$, can be computed directly from Eq. (4.100). On the other hand, given the velocity state of the end effector, the joint rates required to produce that velocity can be computed from Eq. (4.101) without going through numerical inversion of the Jacobian matrix.

4.7.2 Screw-based Jacobian of an Elbow Manipulator

Figure 4.10 shows a schematic diagram of the elbow manipulator studied in Chapter 2. We wish to derive the screw-based Jacobian of this manipulator. Using the coordinate systems established in the figure, the D-H link parameters are listed in Table 4.2, from which the D-H transformation matrices can be developed as follows:

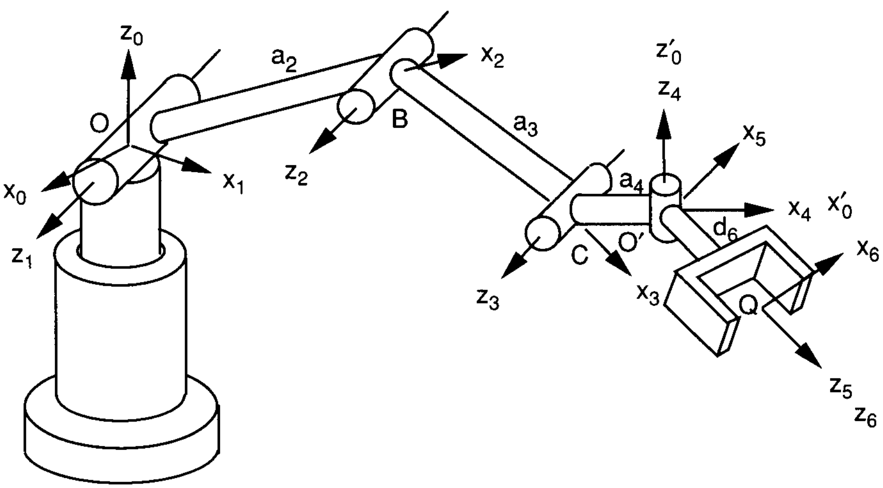

FIGURE 4.10. Elbow manipulator.

TABLE 4.2. Link Parameters of the Elbow Manipulator

Joint i	α_i	a_i	d_i	θ_i
1	$\pi/2$	0	0	variable
2	0	a_2	0	variable
3	0	a_3	0	variable
4	$-\pi/2$	a_4	0	variable
5	$\pi/2$	0	0	variable
6	0	0	d_6	variable

$$
{}^0A_1 = \begin{bmatrix} c\theta_1 & 0 & s\theta_1 & 0 \\ s\theta_1 & 0 & -c\theta_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^1A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2 c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2 s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^2A_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_3 c\theta_3 \\ s\theta_3 & c\theta_3 & 0 & a_3 s\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^3A_4 = \begin{bmatrix} c\theta_4 & 0 & -s\theta_4 & a_4 c\theta_4 \\ s\theta_4 & 0 & c\theta_4 & a_4 s\theta_4 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^4A_5 = \begin{bmatrix} c\theta_5 & 0 & s\theta_5 & 0 \\ s\theta_5 & 0 & -c\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad {}^5A_6 = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ s\theta_6 & c\theta_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

Let an instantaneous reference frame (x_0', y_0', z_0') be aligned with the (x_4, y_4, z_4) link frame as shown in Fig. 4.10. Then the initial conditions for $j = 4$ are $\mathbf{s}_5 = [0, 0, 1]^{\mathrm{T}}$ and $\mathbf{s}_{o,5} = [0, 0, 0]^{\mathrm{T}}$. Applying Eq. (4.83), we obtain the direction and location of the sixth joint axis as follows. For $i = 5$, we obtain

$$
\mathbf{s}_6 = {}^4R_5\, {}^5\mathbf{z}_5 = \begin{bmatrix} s\theta_5 \\ -c\theta_5 \\ 0 \end{bmatrix}, \tag{4.102}
$$

$$
\mathbf{s}_{o,6} = \mathbf{s}_{o,5} + {}^4R_5\, {}^5r_5 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \tag{4.103}
$$

Applying Eq. (4.84), we obtain the directions and locations of the third, second, and first joint axes in sequence as follows. For $i = 3$, we obtain

$$\mathbf{s}_4 = {}^4R_3\,{}^3\mathbf{z}_3 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \tag{4.104}$$

$$\mathbf{s}_{o,4} = \mathbf{s}_{o,5} - {}^4R_4\,{}^4r_4 = \begin{bmatrix} -a_4 \\ 0 \\ 0 \end{bmatrix}, \tag{4.105}$$

$${}^4R_2 = {}^4R_3\,{}^3R_2 = \begin{bmatrix} c\theta_{34} & s\theta_{34} & 0 \\ 0 & 0 & -1 \\ -s\theta_{34} & c\theta_{34} & 0 \end{bmatrix}. \tag{4.106}$$

For $i = 2$, we obtain

$$\mathbf{s}_3 = {}^4R_2\,{}^2\mathbf{z}_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \tag{4.107}$$

$$\mathbf{s}_{o,3} = \mathbf{s}_{o,4} - {}^4R_3\,{}^3r_3 = \begin{bmatrix} -a_3c\theta_4 - a_4 \\ 0 \\ a_3s\theta_4 \end{bmatrix}, \tag{4.108}$$

$${}^4R_1 = {}^4R_2\,{}^2R_1 = \begin{bmatrix} c\theta_{234} & s\theta_{234} & 0 \\ 0 & 0 & -1 \\ -s\theta_{234} & c\theta_{234} & 0 \end{bmatrix}. \tag{4.109}$$

For $i = 1$, we obtain

$$\mathbf{s}_2 = {}^4R_1\,{}^1\mathbf{z}_1 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \tag{4.110}$$

$$\mathbf{s}_{o,2} = \mathbf{s}_{o,3} - {}^4R_2\,{}^2r_2 = \begin{bmatrix} -a_2c\theta_{34} - a_3c\theta_4 - a_4 \\ 0 \\ a_2s\theta_{34} + a_3s\theta_4 \end{bmatrix}, \tag{4.111}$$

$${}^4R_0 = {}^4R_1\,{}^1R_0 = \begin{bmatrix} c\theta_1c\theta_{234} & s\theta_1c\theta_{234} & s\theta_{234} \\ -s\theta_1 & c\theta_1 & 0 \\ -c\theta_1s\theta_{234} & -s\theta_1s\theta_{234} & c\theta_{234} \end{bmatrix}. \tag{4.112}$$

For $i = 0$, we obtain

$$\mathbf{s}_1 = {}^4R_0\,{}^0\mathbf{z}_0 = \begin{bmatrix} s\theta_{234} \\ 0 \\ c\theta_{234} \end{bmatrix}, \tag{4.113}$$

$$\mathbf{s}_{o,1} = \mathbf{s}_{o,2} - {}^4R_1\,{}^1r_1 = \begin{bmatrix} -a_2 c\theta_{34} - a_3 c\theta_4 - a_4 \\ 0 \\ a_2 s\theta_{34} + a_3 s\theta_4 \end{bmatrix}. \tag{4.114}$$

We now apply Eq. (4.49) to compute the Jacobian matrix column by column:

$${}^4J = \begin{bmatrix} s\theta_{234} & 0 & 0 & 0 & 0 & s\theta_5 \\ 0 & -1 & -1 & -1 & 0 & -c\theta_5 \\ c\theta_{234} & 0 & 0 & 0 & 1 & 0 \\ 0 & a_2 s\theta_{34} + a_3 s\theta_4 & a_3 s\theta_4 & 0 & 0 & 0 \\ x_{51} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_2 c\theta_{34} + a_3 c\theta_4 + a_4 & a_3 c\theta_4 + a_4 & a_4 & 0 & 0 \end{bmatrix}, \tag{4.115}$$

where $x_{51} = a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234}$. Equation (4.115) relates the joint rates to the velocity state of the end effector. The leading superscript, 4, indicates that the joint screws, $\$_i$, $i = 1, 2, \ldots, n$, are expressed in a reference frame that is instantaneously coincident with the fourth link frame. Substituting Eq. (4.115) into (4.85), we obtain

$$\begin{aligned} {}^4\omega_x &= \dot\theta_1 s\theta_{234} + \dot\theta_6 s\theta_5, \\ {}^4\omega_y &= -\dot\theta_{234} - \dot\theta_6 c\theta_5, \\ {}^4\omega_z &= \dot\theta_1 c\theta_{234} + \dot\theta_5, \\ {}^4v_{ox} &= \dot\theta_2 a_2 s\theta_{34} + \dot\theta_{23} a_3 s\theta_4, \\ {}^4v_{oy} &= \dot\theta_1 (a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234}), \\ {}^4v_{oz} &= \dot\theta_2 a_2 c\theta_{34} + \dot\theta_{23} a_3 c\theta_4 + \dot\theta_{234} a_4. \end{aligned} \tag{4.116}$$

Hence, given the joint rates, the end-effector velocity state can be computed directly from Eq. (4.116). On the other hand, given the velocity state of the end effector, the joint rates can be found in sequence by the following inverse transformation:

$$\begin{aligned} \dot\theta_1 &= \frac{{}^4v_{oy}}{a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234}}, \\ \dot\theta_5 &= {}^4\omega_z - \dot\theta_1 c\theta_{234}, \\ \dot\theta_6 &= \frac{{}^4\omega_x - \dot\theta_1 s\theta_{234}}{s\theta_5}, \\ \dot\theta_{234} &= -{}^4\omega_y - \dot\theta_6 c\theta_5, \end{aligned}$$

$$\dot{\theta}_{23} = \frac{s\theta_{34}({}^4v_{oz} - a_4\dot{\theta}_{234}) - c\theta_{34}\,{}^4v_{ox}}{a_3 s\theta_3},$$
$$\dot{\theta}_2 = \frac{c\theta_4\,{}^4v_{ox} - s\theta_4({}^4v_{oz} - a_4\dot{\theta}_{234})}{a_2 s\theta_3},$$
$$\dot{\theta}_3 = \dot{\theta}_{23} - \dot{\theta}_2,$$
$$\dot{\theta}_4 = \dot{\theta}_{234} - \dot{\theta}_{23}. \tag{4.117}$$

4.7.3 Screw-based Jacobian of a Nearly General 6*R* Manipulator

In previous examples it was possible to derive analytical inversions of the Jacobian matrices due to the special geometric arrangement of the links. The complexity of the Jacobian increases rapidly as the geometry of a manipulator becomes more general. However, substantial simplification can still be achieved by choosing a proper reference frame. In this example, to illustrate the point, we derive the screw-based Jacobian of a nearly general 6-dof, 6*R* manipulator as shown in Fig. 4.11.

For this manipulator, the first and second joint axes intersect perpendicularly; the second and third joint axes are parallel to each other; and the third and fourth joint axes intersect perpendicularly, as do the fourth and fifth joint axes. The sixth joint axis is perpendicular to the fifth with an offset distance a_5. The D-H link parameters are summarized in Table 4.3.

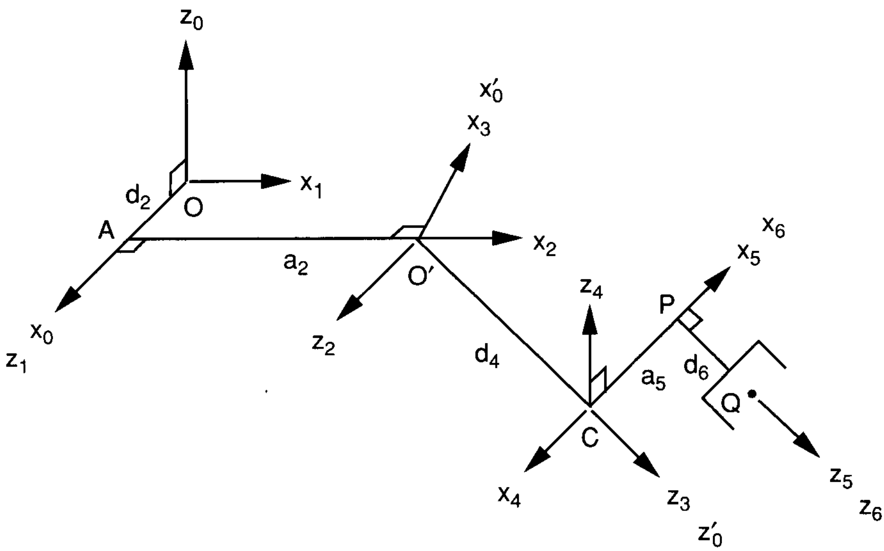

FIGURE 4.11. Nearly general manipulator.

TABLE 4.3. Link Parameters of a Nearly General Manipulator

Joint i	α_i	a_i	d_i	θ_i
1	$\pi/2$	0	0	variable
2	0	a_2	d_2	variable
3	$\pi/2$	0	0	variable
4	$\pi/2$	0	d_4	variable
5	$\pi/2$	a_5	0	variable
6	0	0	d_6	variable

Hence the D-H transformation matrices can be written as

$$
{}^0A_1 = \begin{bmatrix} c\theta_1 & 0 & s\theta_1 & 0 \\ s\theta_1 & 0 & -c\theta_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
{}^1A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2s\theta_2 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^2A_3 = \begin{bmatrix} c\theta_3 & 0 & s\theta_3 & 0 \\ s\theta_3 & 0 & -c\theta_3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
{}^3A_4 = \begin{bmatrix} c\theta_4 & 0 & s\theta_4 & 0 \\ s\theta_4 & 0 & -c\theta_4 & 0 \\ 0 & 1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
{}^4A_5 = \begin{bmatrix} c\theta_5 & 0 & s\theta_5 & a_5c\theta_2 \\ s\theta_5 & 0 & -c\theta_5 & a_5s\theta_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
{}^5A_6 = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ s\theta_6 & c\theta_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

Let the instantaneous reference frame (x_0', y_0', z_0') be aligned with the (x_3, y_3, z_3) link frame. Then the initial conditions for $j = 3$ are $\mathbf{s}_4 = [0, 0, 1]^T$ and $\mathbf{s}_{o,4} = [0, 0, 0]^T$. Applying Eq. (4.83), we obtain the directions and locations of the fifth and sixth joint axes as follows. For $i = 4$, we obtain

$$
\mathbf{s}_5 = {}^3R_4\,{}^4\mathbf{z}_4 = \begin{bmatrix} s\theta_4 \\ -c\theta_4 \\ 0 \end{bmatrix}, \tag{4.118}
$$

$$
\mathbf{s}_{o,5} = \mathbf{s}_{o,4} + {}^3R_4\,{}^4r_4 = \begin{bmatrix} 0 \\ 0 \\ d_4 \end{bmatrix}, \tag{4.119}
$$

$$
{}^3R_5 = {}^3R_4\,{}^4R_5 = \begin{bmatrix} c\theta_4c\theta_5 & s\theta_4 & c\theta_4s\theta_5 \\ s\theta_4c\theta_5 & -c\theta_4 & s\theta_4s\theta_5 \\ s\theta_5 & 0 & -c\theta_5 \end{bmatrix}. \tag{4.120}
$$

For $i = 5$, we obtain

$$\mathbf{s}_6 = {}^3R_5\,{}^5\mathbf{z}_5 = \begin{bmatrix} c\theta_4 s\theta_5 \\ s\theta_4 s\theta_5 \\ -c\theta_5 \end{bmatrix}, \tag{4.121}$$

$$\mathbf{s}_{o,6} = \mathbf{s}_{o,5} + {}^3R_5\,{}^5r_5 = \begin{bmatrix} a_5 c\theta_4 c\theta_5 \\ a_5 s\theta_4 c\theta_5 \\ d_4 + a_5 s\theta_5 \end{bmatrix}. \tag{4.122}$$

Applying Eq. (4.84), we obtain the directions and locations of the third, second, and first joint axes as follows. For $i = 2$, we obtain

$$\mathbf{s}_3 = {}^3R_2\,{}^2\mathbf{z}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \tag{4.123}$$

$$\mathbf{s}_{o,3} = \mathbf{s}_{o,4} - {}^3R_3\,{}^3r_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \tag{4.124}$$

$${}^3R_1 = {}^3R_2\,{}^2R_1 = \begin{bmatrix} c\theta_{23} & s\theta_{23} & 0 \\ 0 & 0 & 1 \\ s\theta_{23} & -c\theta_{23} & 0 \end{bmatrix}. \tag{4.125}$$

For $i = 1$, we obtain

$$\mathbf{s}_2 = {}^3R_1\,{}^1\mathbf{z}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \tag{4.126}$$

$$\mathbf{s}_{o,2} = \mathbf{s}_{o,3} - {}^3R_2\,{}^2r_2 = \begin{bmatrix} -a_2 c\theta_3 \\ -d_2 \\ -a_2 s\theta_3 \end{bmatrix}, \tag{4.127}$$

$${}^3R_0 = {}^3R_1\,{}^1R_0 = \begin{bmatrix} c\theta_1 c\theta_{23} & s\theta_1 c\theta_{23} & s\theta_{23} \\ s\theta_1 & -c\theta_1 & 0 \\ c\theta_1 s\theta_{23} & s\theta_1 s\theta_{23} & -c\theta_{23} \end{bmatrix}. \tag{4.128}$$

For $i = 0$, we obtain

$$\mathbf{s}_1 = {}^3R_0\,{}^0\mathbf{z}_0 = \begin{bmatrix} s\theta_{23} \\ 0 \\ -c\theta_{23} \end{bmatrix}, \tag{4.129}$$

$$\mathbf{s}_{o,1} = \mathbf{s}_{o,2} - {}^3R_1\,{}^1r_1 = \begin{bmatrix} -a_2 c\theta_3 \\ -d_2 \\ -a_2 s\theta_3 \end{bmatrix}. \tag{4.130}$$

We now apply Eqs. (4.49) and (4.50) to compute the Jacobian matrix column by column. As a result, we obtain

$${}^3J = \begin{bmatrix} s\theta_{23} & 0 & 0 & 0 & s\theta_4 & c\theta_4 s\theta_5 \\ 0 & 1 & 1 & 0 & -c\theta_4 & s\theta_4 s\theta_5 \\ -c\theta_{23} & 0 & 0 & 1 & 0 & -c\theta_5 \\ d_2 c\theta_{23} & a_2 s\theta_3 & 0 & 0 & d_4 c\theta_4 & -s\theta_4(a_5 + d_4 s\theta_5) \\ -a_2 c\theta_2 & 0 & 0 & 0 & d_4 s\theta_4 & c\theta_4(a_5 + d_4 s\theta_5) \\ d_2 s\theta_{23} & -a_2 c\theta_3 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{4.131}$$

We observe that the Jacobian is still simple enough to make an analytical inversion feasible.

4.8 TRANSFORMATION OF SCREW COORDINATES

In the preceding section we have shown that the Jacobian matrix can be greatly simplified by using a reference frame that is instantaneously coincident with an intermediate link frame of the manipulator. In practice, however, the velocity state of the end effector is often specified in a fixed reference frame (x_0, y_0, z_0). Therefore, it is necessary to transform the velocity state from the fixed reference frame to the instantaneous reference frame such that the corresponding joint rates can be computed. In what follows, a 6×6 matrix of transformation, $\tilde{T}$, is derived to serve this purpose (Yuan et al., 1971).

As shown in Fig. 4.12, let (x_i, y_i, z_i) and (x_j, y_j, z_j) be two reference frames of interest. The position of O_j relative to the (x_i, y_i, z_i) frame is given by ${}^i\mathbf{p} = [p_x, p_y, p_z]^{\mathrm{T}}$, and the orientation of the (x_j, y_j, z_j) frame relative to the (x_i, y_i, z_i) frame is described by a rotation matrix ${}^iR_j$. A screw \$ relative the (x_i, y_i, z_i) frame is denoted by ${}^i\$$, and the same screw relative to the (x_j, y_j, z_j) frame is denoted by ${}^j\$$.

Following the definition of a screw, we have

$${}^i\$ = \begin{bmatrix} {}^i\mathbf{s} \\ {}^i\mathbf{s}_o \times {}^i\mathbf{s} + \lambda\,{}^i\mathbf{s} \end{bmatrix} \tag{4.132}$$

$${}^j\$ = \begin{bmatrix} {}^j\mathbf{s} \\ {}^j\mathbf{s}_o \times {}^j\mathbf{s} + \lambda\,{}^j\mathbf{s} \end{bmatrix}. \tag{4.133}$$

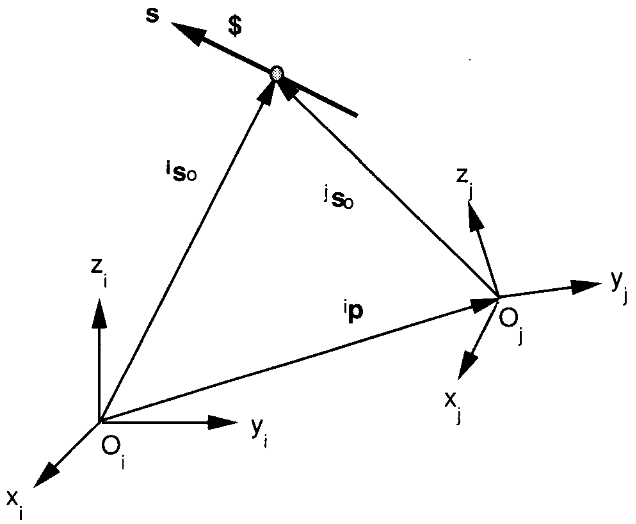

FIGURE 4.12. Coordinate transformation of a screw.

However, the line and moment vectors of the two screws are related by the following transformations:

$$^i\mathbf{s} = {}^iR_j\,{}^j\mathbf{s}, \tag{4.134}$$

$$^i\mathbf{s}_o = {}^i\mathbf{p} + {}^iR_j\,{}^j\mathbf{s}_o. \tag{4.135}$$

Hence

$$^i\mathbf{s}_o \times {}^i\mathbf{s} = ({}^i\mathbf{p} + {}^iR_j\,{}^j\mathbf{s}_o) \times {}^i\mathbf{s} = {}^i\mathbf{p} \times ({}^iR_j\,{}^j\mathbf{s}) + {}^iR_j({}^j\mathbf{s}_o \times {}^j\mathbf{s}). \tag{4.136}$$

Substituting Eqs. (4.134) through (4.136) into (4.132), we obtain

$$^i\$ = \begin{bmatrix} {}^iR_j\,{}^j\mathbf{s} \\ {}^iR_j({}^j\mathbf{s}_o \times {}^j\mathbf{s}) + {}^i\mathbf{p} \times ({}^iR_j\,{}^j\mathbf{s}) + \lambda\,{}^iR_j\,{}^j\mathbf{s} \end{bmatrix}. \tag{4.137}$$

Equation (4.137) can be written in the following matrix form:

$$^i\$ = {}^i\tilde{T}_j\,{}^j\$, \tag{4.138}$$

where

$$^i\tilde{T}_j = \left[\begin{array}{c:c} {}^iR_j & \mathbf{0} \\ \hdashline {}^iW_j\,{}^iR_j & {}^iR_j \end{array}\right] \tag{4.139}$$

is a 6×6 matrix, and

$$^{i}W_j = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix} \tag{4.140}$$

is a 3×3 skew-symmetric matrix representing the vector of $\overline{O_i O_j}$ expressed in the ith frame.

Since $^{i}W_j$ is skew symmetric and $^{i}R_j$ is orthogonal, the inverse transformation matrix can be written as

$$^{j}\tilde{T}_i = \left[\begin{array}{c:c} {}^{i}R_j^{\mathrm{T}} & \mathbf{0} \\ \hdashline {}^{i}R_j^{\mathrm{T}}\,{}^{i}W_j^{\mathrm{T}} & {}^{i}R_j^{\mathrm{T}} \end{array}\right]. \tag{4.141}$$

Hence given a screw in the jth frame, we can express it in the ith frame by applying Eq. (4.139), and vice versa using Eq. (4.141).

Geometrically, the six columns of $^{i}\tilde{T}_j$ in Eq. (4.139) represent three normalized screws of zero pitch along and three normalized screws of infinite pitch parallel to the coordinate axes of the (x_j, y_j, z_j) frame and expressed in the (x_i, y_i, z_i) frame, respectively. These six normalized screws, expressed in the (x_j, y_j, z_j) frame, are

$$\begin{aligned} \hat{\$}_1 &= [1, 0, 0, 0, 0, 0]^{\mathrm{T}}, \\ \hat{\$}_2 &= [0, 1, 0, 0, 0, 0]^{\mathrm{T}}, \\ \hat{\$}_3 &= [0, 0, 1, 0, 0, 0]^{\mathrm{T}}, \\ \hat{\$}_4 &= [0, 0, 0, 1, 0, 0]^{\mathrm{T}}, \\ \hat{\$}_5 &= [0, 0, 0, 0, 1, 0]^{\mathrm{T}}, \\ \hat{\$}_6 &= [0, 0, 0, 0, 0, 1]^{\mathrm{T}}. \end{aligned} \tag{4.142}$$

The geometric interpretation above can sometimes be very helpful in deriving the transformation matrix. In addition, it can also be used as a check on the validity of any analytical derivation of $\tilde{T}$.

Example 4.8.1 *Stanford Manipulator* Let us consider the Stanford manipulator shown in Fig. 4.8 as an example to illustrate the foregoing principle. Using the D-H link parameters listed in the preceding section, we obtain the matrix product $^{0}A_3$ as

$$
{}^0A_3 = \begin{bmatrix} s\theta_1 & c\theta_1 c\theta_2 & c\theta_1 s\theta_2 & d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 \\ -c\theta_1 & s\theta_1 c\theta_2 & s\theta_1 s\theta_2 & d_3 s\theta_1 s\theta_2 + d_2 c\theta_1 \\ 0 & -s\theta_2 & c\theta_2 & d_3 c\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4.143}
$$

Hence the rotation matrix ${}^0R_3$ is given by

$$
{}^0R_3 = \begin{bmatrix} s\theta_1 & c\theta_1 c\theta_2 & c\theta_1 s\theta_2 \\ -c\theta_1 & s\theta_1 c\theta_2 & s\theta_1 s\theta_2 \\ 0 & -s\theta_2 & c\theta_2 \end{bmatrix}. \tag{4.144}
$$

The position vector of O_3 with respect to the fixed reference frame (x_0, y_0, z_0) is given by

$$
{}^0\mathbf{p}_3 = {}^0A_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} d_3 c\theta_1 s\theta_2 - d_2 s\theta_1 \\ d_3 s\theta_1 s\theta_2 + d_2 c\theta_1 \\ d_3 c\theta_2 \\ 1 \end{bmatrix}. \tag{4.145}
$$

Substituting Eqs. (4.144) and (4.145) into (4.139), we obtain

$$
{}^0\tilde{T}_3 =
\begin{bmatrix} s\theta_1 & c\theta_1 c\theta_2 & c\theta_1 s\theta_2 & 0 & 0 & 0 \\ -c\theta_1 & s\theta_1 c\theta_2 & s\theta_1 s\theta_2 & 0 & 0 & 0 \\ 0 & -s\theta_2 & c\theta_2 & 0 & 0 & 0 \\ d_3 c\theta_1 c\theta_2 & -d_3 s\theta_1 - d_2 c\theta_1 s\theta_2 & d_2 c\theta_1 c\theta_2 & s\theta_1 & c\theta_1 c\theta_2 & c\theta_1 s\theta_2 \\ d_3 s\theta_1 c\theta_2 & d_3 c\theta_1 - d_2 s\theta_1 s\theta_2 & d_2 s\theta_1 c\theta_2 & -c\theta_1 & s\theta_1 c\theta_2 & s\theta_1 s\theta_2 \\ -d_3 s\theta_2 & -d_2 c\theta_2 & -d_2 s\theta_2 & 0 & -s\theta_2 & c\theta_2 \end{bmatrix}. \tag{4.146}
$$

The inverse transformation is given by

$$
{}^3\tilde{T}_0 =
\begin{bmatrix} s\theta_1 & -c\theta_1 & 0 & 0 & 0 & 0 \\ c\theta_1 c\theta_2 & s\theta_1 c\theta_2 & -s\theta_2 & 0 & 0 & 0 \\ c\theta_1 s\theta_2 & s\theta_1 s\theta_2 & c\theta_2 & 0 & 0 & 0 \\ d_3 c\theta_1 c\theta_2 & d_3 s\theta_1 c\theta_2 & -d_3 s\theta_2 & s\theta_1 & -c\theta_1 & 0 \\ -d_3 s\theta_1 - d_2 c\theta_1 s\theta_2 & d_3 c\theta_1 - d_2 s\theta_1 s\theta_2 & -d_2 c\theta_2 & c\theta_1 c\theta_2 & s\theta_1 c\theta_2 & -s\theta_2 \\ d_2 c\theta_1 c\theta_2 & d_2 s\theta_1 c\theta_2 & -d_2 s\theta_2 & c\theta_1 s\theta_2 & s\theta_1 s\theta_2 & c\theta_2 \end{bmatrix}. \tag{4.147}
$$

In a path planning problem, the velocity state of the end effector is often specified by the angular velocity vector, $(\omega_x, \omega_y, \omega_z)$, and the velocity of a point in the end effector, (v_{ox}, v_{oy}, v_{oz}), which is instantaneously coincident with the fixed origin O_0. These six components are precisely the coordinates of a screw about which the end effector is twisting instantaneously. Consequently, these six components are related to the components of the velocity state specified in the third moving frame by the following transformation.

$$\begin{bmatrix} {}^3\omega_x \\ {}^3\omega_y \\ {}^3\omega_z \\ {}^3v_{ox} \\ {}^3v_{oy} \\ {}^3v_{oz} \end{bmatrix} = {}^3\tilde{T}_0 \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \\ v_{ox} \\ v_{oy} \\ v_{oz} \end{bmatrix}. \tag{4.148}$$

Once the velocity state is transformed into the instantaneous reference frame, the inverse velocity problem, namely the problem of finding the joint rates needed to achieve a given velocity state, can be solved by

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{d}_3 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \\ \dot{\theta}_6 \end{bmatrix} = {}^3J^{-1} \begin{bmatrix} {}^3\omega_x \\ {}^3\omega_y \\ {}^3\omega_z \\ {}^3v_{ox} \\ {}^3v_{oy} \\ {}^3v_{oz} \end{bmatrix}. \tag{4.149}$$

Example 4.8.2 *Elbow Manipulator* Consider the elbow manipulator shown in Fig. 4.10 as a second example. Using the D-H link parameters listed in the preceding section, we obtain the matrix product ${}^0A_4$ as

$${}^0A_4 = \begin{bmatrix} c\theta_1 c\theta_{234} & -s\theta_1 & -c\theta_1 s\theta_{234} & c\theta_1(a_4 c\theta_{234} + a_3 c\theta_{23} + a_2 c\theta_2) \\ s\theta_1 c\theta_{234} & c\theta_1 & -s\theta_1 s\theta_{234} & s\theta_1(a_4 c\theta_{234} + a_3 c\theta_{23} + a_2 c\theta_2) \\ s\theta_{234} & 0 & c\theta_{234} & a_4 s\theta_{234} + a_3 s\theta_{23} + a_2 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{4.150}$$

Hence the rotation matrix ${}^0R_4$ is given by

$${}^0R_4 = \begin{bmatrix} c\theta_1 c\theta_{234} & -s\theta_1 & -c\theta_1 s\theta_{234} \\ s\theta_1 c\theta_{234} & c\theta_1 & -s\theta_1 s\theta_{234} \\ s\theta_{234} & 0 & c\theta_{234} \end{bmatrix}. \tag{4.151}$$

The position vector of O_4 with respect to the fixed reference frame (x_0, y_0, z_0) is given by

$$
{}^0\mathbf{p}_4 = {}^0A_4 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} c\theta_1(a_4c\theta_{234} + a_3c\theta_{23} + a_2c\theta_2) \\ s\theta_1(a_4c\theta_{234} + a_3c\theta_{23} + a_2c\theta_2) \\ a_4s\theta_{234} + a_3s\theta_{23} + a_2s\theta_2 \\ 1 \end{bmatrix}. \tag{4.152}
$$

Substituting Eqs. (4.151) and (4.152) into (4.139), we obtain

$$
{}^0\tilde{T}_4 = \begin{bmatrix} c\theta_1c\theta_{234} & -s\theta_1 & -c\theta_1s\theta_{234} & 0 & 0 & 0 \\ s\theta_1c\theta_{234} & c\theta_1 & -s\theta_1s\theta_{234} & 0 & 0 & 0 \\ s\theta_{234} & 0 & c\theta_{234} & 0 & 0 & 0 \\ x_1s\theta_1 & -x_2c\theta_1 & x_4s\theta_1 & c\theta_1c\theta_{234} & -s\theta_1 & -c\theta_1s\theta_{234} \\ -x_1c\theta_1 & -x_2s\theta_1 & -x_4c\theta_1 & s\theta_1c\theta_{234} & c\theta_1 & -s\theta_1s\theta_{234} \\ 0 & x_3 & 0 & s\theta_{234} & 0 & c\theta_{234} \end{bmatrix}, \tag{4.153}
$$

where $x_1 = a_3s\theta_4 + a_2s\theta_{34}$, $x_2 = a_4s\theta_{234} + a_3s\theta_{23} + a_2s\theta_2$, $x_3 = a_4c\theta_{234} + a_3c\theta_{23} + a_2c\theta_2$, and $x_4 = a_4 + a_3c\theta_4 + a_2c\theta_{34}$. The inverse transformation is given by

$$
{}^4\tilde{T}_0 = \begin{bmatrix} c\theta_1c\theta_{234} & s\theta_1c\theta_{234} & s\theta_{234} & 0 & 0 & 0 \\ -s\theta_1 & c\theta_1 & 0 & 0 & 0 & 0 \\ -c\theta_1s\theta_{234} & -s\theta_1s\theta_{234} & c\theta_{234} & 0 & 0 & 0 \\ x_1s\theta_1 & -x_1c\theta_1 & 0 & c\theta_1c\theta_{234} & s\theta_1c\theta_{234} & s\theta_{234} \\ -x_2c\theta_1 & -x_2s\theta_1 & x_3 & -s\theta_1 & c\theta_1 & 0 \\ x_4s\theta_1 & -x_4c\theta_1 & 0 & -c\theta_1s\theta_{234} & -s\theta_1s\theta_{234} & c\theta_{234} \end{bmatrix}. \tag{4.154}
$$

Hence the velocity state of the end effector with respect to the fixed reference frame is related to that with respect to the fourth moving frame by the following matrix transformation:

$$
\begin{bmatrix} {}^4\omega_x \\ {}^4\omega_y \\ {}^4\omega_z \\ {}^4v_{ox} \\ {}^4v_{oy} \\ {}^4v_{oz} \end{bmatrix} = {}^4\tilde{T}_0 \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \\ v_{ox} \\ v_{oy} \\ v_{oz} \end{bmatrix}. \tag{4.155}
$$

Similar to the Stanford manipulator, the inverse velocity problem can be solved as

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \\ \dot{\theta}_6 \end{bmatrix} = {}^4J^{-1} \begin{bmatrix} {}^4\omega_x \\ {}^4\omega_y \\ {}^4\omega_z \\ {}^4v_{ox} \\ {}^4v_{oy} \\ {}^4v_{oz} \end{bmatrix}. \tag{4.156}$$

4.9 RELATIONSHIP BETWEEN THE TWO METHODS

The velocity vectors, $\boldsymbol{\omega}_n$ and $\mathbf{v}_n$, used by the conventional method are expressed in the fixed reference frame, while ${}^j\boldsymbol{\omega}_n$ and ${}^j\mathbf{v}_o$ employed by the method of screw coordinates are expressed in the instantaneous reference frame. These two velocity states are related by the following transformation:

$$\boldsymbol{\omega}_n = {}^0R_j\, {}^j\boldsymbol{\omega}_n \tag{4.157}$$

$$\mathbf{v}_n = {}^0R_j({}^j\mathbf{v}_o + {}^j\boldsymbol{\omega}_n \times {}^j\mathbf{p}_n) \tag{4.158}$$

where

$$\begin{bmatrix} {}^j\mathbf{p}_n \\ 1 \end{bmatrix} = {}^jA_n \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

denotes the position vector of the origin of the end-effector frame with respect to the jth link frame.

4.10 CONDITION NUMBER

The Jacobian matrix J transforms the joint rate in n-dimensional space into the end-effector velocity in m-dimensional space. In a conventional Jacobian, the first three elements of $\dot{\mathbf{x}}$ have the dimension of length per unit time, whereas the last three elements of $\dot{\mathbf{x}}$ have the dimension of radians per unit time. Further, depending on the type of joints used in a manipulator, the elements of $\dot{\mathbf{q}}$ may not necessarily have the same dimension. As a result, the elements of the Jacobian matrix do not necessarily have uniform dimensions. For example, if all the joints are revolute, the first three rows of the Jacobian matrix J have the dimension of length, whereas the last three rows are dimensionless. Therefore, one should be careful in manipulating the matrix.

For those manipulators with only one type of joint and for one type of task, namely, either point positioning or body orienting but not both, the Jacobian matrix can be characterized by a measure called the *condition number*, c. The condition number of a matrix A is defined as

$$c = \|A\|\,\|A^{-1}\|, \tag{4.159}$$

where the *norm* of A is defined as

$$\|A\| = \max_{x \neq 0} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|}.$$

In other words, the norm of A bounds the amplifying power of the matrix:

$$\|A\mathbf{x}\| \le \|A\|\,\|\mathbf{x}\| \qquad \text{for all vectors } \mathbf{x}, \tag{4.160}$$

where the equality holds for at least one nonzero $\mathbf{x}$ (Strang, 1988). In case A is a positive-definite matrix, the condition number of A becomes the ratio of the largest eigenvalue to the smallest eigenvalue of A.

The condition number of the Jacobian matrix depends on the link lengths and the manipulator configuration. As the end effector moves from location to location, the condition number will assume different values. The minimum condition number of any matrix is 1. Those points in the workspace of a manipulator where the condition number of the Jacobian matrix is equal to 1 are called *isotropic points* (Salisbury, 1982). Depending on link dimensions, a manipulator may or may not possess isotropic points within its workspace.

Another method of comparing the transformation characteristics is to compare the joint rates required to produce a unity end-effector velocity in all possible directions. To achieve this goal, we confine the end-effector velocity vector on an m-dimensional unit sphere,

$$\dot{\mathbf{x}}^{\mathrm{T}}\dot{\mathbf{x}} = 1 \tag{4.161}$$

and compare the corresponding joint rates in the n-dimensional joint space. Substituting Eq. (4.56) into (4.161) yields

$$\dot{\mathbf{q}}^{\mathrm{T}} J^{\mathrm{T}} J \dot{\mathbf{q}} = 1. \tag{4.162}$$

Equation (4.162) represents an ellipsoid in the n-dimensional joint space. Because the product $J^{\mathrm{T}}J$ is symmetric positive semidefinite, its eigenvectors are orthogonal. The principal axes of the ellipsoid coincide with the eigenvectors of $J^{\mathrm{T}}J$, and the lengths of its principal axes are equal to the reciprocals of the square roots of the eigenvalues of $J^{\mathrm{T}}J$.

Since the Jacobian matrix is configuration dependent, the ellipsoid is also configuration dependent. As the end effector moves from one location to another, the shape and orientation of the ellipsoid will also change accordingly. The closer the velocity ellipsoid to a sphere, the better the transformation characteristics are. The transformation is said to be *isotropic* when the principal axes are all of equal length. At an isotropic point, a unit sphere in the m-dimensional end-effector space maps onto a sphere in the n-dimensional joint space. On the other hand, at a singular point, one or more of the principal axes becomes infinitely long and the ellipsoid degenerates into a cylinder. Under such a condition, the end effector will not be able to move in some directions.

Example 4.10.1 *Planar 2-DOF Manipulator* Consider the planar 2-dof manipulator shown in Fig. 4.7 as an example. Let $a_1 = \sqrt{2}$ m and $a_2 = 1$ m. Then the Jacobian matrix, Eq. (4.66), can be written as

$$J = \begin{bmatrix} -\sqrt{2}s\theta_1 - s\theta_{12} & -s\theta_{12} \\ \sqrt{2}c\theta_1 + c\theta_{12} & c\theta_{12} \end{bmatrix} \quad \text{m}. \tag{4.163}$$

The matrix product J^TJ is given by

$$J^TJ = \begin{bmatrix} 2\sqrt{2}c\theta_2 + 3 & \sqrt{2}c\theta_2 + 1 \\ \sqrt{2}c\theta_2 + 1 & 1 \end{bmatrix} \quad \text{m}^2. \tag{4.164}$$

Note that the matrix product J^TJ is symmetric and independent of θ_1. The eigenvalues of J^TJ are $\lambda_1 = (2-\sqrt{2})(-c\theta_2+1)$ and $\lambda_2 = (2+\sqrt{2})(c\theta_2+1)$, respectively. In particular, $\lambda_1 = \lambda_2 = 1$ for the configuration of $\theta_2 = 3\pi/4$ or $\theta_2 = 5\pi/4$. We conclude that the manipulator possesses a unity condition number when it assumes either one of these two configurations. Since the condition number is independent of θ_1, all points swept by the manipulator with $\theta_2 = 3\pi/4$ or $\theta_2 = 5\pi/4$ form the locus of isotropic points as shown in Fig. 4.13.

For the purpose of discussion, let us further assume that $\theta_2 = \pi/2$. Then the matrix product J^TJ becomes

$$J^TJ = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}. \tag{4.165}$$

The eigenvalues of J^TJ are $\lambda_1 = 2 - \sqrt{2} = 0.5858$ and $\lambda_2 = 2 + \sqrt{2} = 3.4142$. The corresponding eigenvectors, normalized to unit length, are $(-0.3827, 0.9238)$ and $(0.9238, 0.3827)$, respectively. These two eigen-

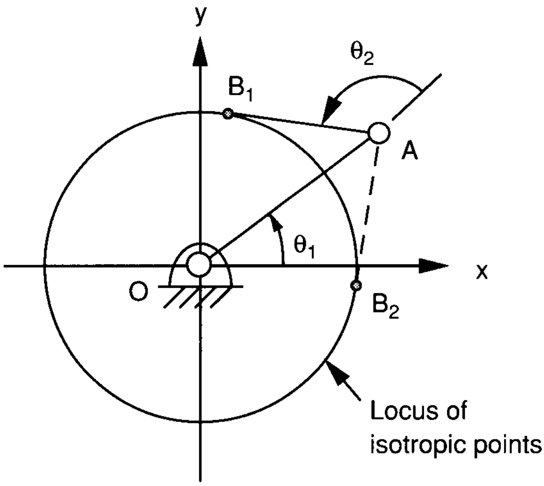

FIGURE 4.13. Locus of isotropic points.

vectors are at 22.5° angles with the $\dot{\theta}_2$ and $\dot{\theta}_1$ axes, respectively, and they are lined up with the principal axes of the ellipse.

Substituting Eq. (4.165) into (4.162) yields

$$\begin{aligned} 3\dot{\theta}_1^2 + 2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2 &= \frac{(0.3827\dot{\theta}_1 - 0.9238\dot{\theta}_2)^2}{1.3065^2} + \frac{(0.9238\dot{\theta}_1 + 0.3827\dot{\theta}_2)^2}{0.5412^2} \\ &= 1. \end{aligned} \tag{4.166}$$

Equation (4.166) represents an ellipse as shown in Fig. 4.14*b*. The joint rates required to produce a unity end-effector velocity are $(\dot{\theta}_1, \dot{\theta}_2) = (-0.500, 1.207)$ rad/s along the major axis, and $(\dot{\theta}_1, \dot{\theta}_2) = (0.500, 0.207)$ rad/s along the minor axis.

Without loss of generality, we assume that $\theta_1 = 0$. Then the Jacobian matrix becomes

$$J = \begin{bmatrix} -1 & -1 \\ \sqrt{2} & 0 \end{bmatrix}. \tag{4.167}$$

Hence the corresponding end-effector velocities are $(v_x, v_y) = (-0, 707, -0.707)$ m/s along the major axis and $(v_x, v_y) = (-0, 707, 0.707)$ m/s along the minor axis, respectively, as shown in Fig. 4.14*a*. We notice that to produce the same end-effector speed along the principal axes, one requires the largest joint rates while the other requires the smallest joint rates.

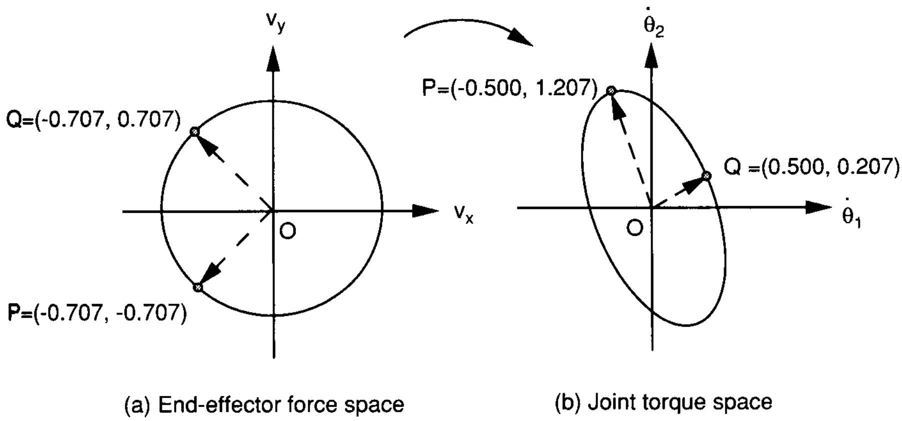

FIGURE 4.14. Velocity ellipsoid.

4.11 SINGULARITY ANALYSIS

The Jacobian matrix, J, transforms the joint rates of a manipulator into the end-effector velocity state. Thus, given the joint rates, we can compute the end-effector velocities directly. In a trajectory planning problem, however, the end-effector velocities are usually given along a desired path in the end-effector space, and these velocities must be converted into the joint rates in the joint space. This requires a computation of the inverse transformation of Eq. (4.56):

$$\dot{\mathbf{q}} = J^{-1}\dot{\mathbf{x}}. \tag{4.168}$$

The inverse transformation matrix, J^{-1}, can be derived by formulating the matrix of cofactors of J, transporting it, and dividing throughout by the determinant of J. Equation (4.168) provides a means of calculating the joint rates required to produce certain desired end-effector velocities. It is obvious that the required joint rates depend on the condition of the Jacobian matrix. At certain manipulator configurations, the Jacobian matrix may lose its full rank (i.e., there is a reduction of the number of linearly independent rows or columns). Hence as the manipulator approaches these configurations, the Jacobian matrix becomes ill conditioned and may not be invertible. Under such a condition, numerical solution of Eq. (4.168) results in infinite joint rates.

A manipulator is said to be at a *singular configuration* when the Jacobian matrix losses its full rank. Physically, this implies that the instantaneous screws spanning the n-dimensional space of the Jacobian matrix become linearly dependent. Therefore, at a singular configuration, a serial manipulator may lose one or more degrees of freedom, and it won't be able to move in some directions in the end-effector space.

Singular configurations can be found by setting the determinant of the Jacobian matrix to zero. In general, this will result in a single algebraic equation. For serial manipulators, the singular condition is a function of the intermediate joint variables, not of the first and last joint variables. This is because the presence of a singularity depends solely on the relative locations of the joint axes. Rotation of the entire manipulator about the first axis does not change the relative locations of the joint axes. Similarly, rotation of the end effector about the last joint axis does not affect the location of any joint axis. Therefore, the first and last joint variables do not appear in the determinant of the Jacobian matrix. Several studies of singularity analysis can be found in the literature (Hunt, 1986, 1987a; Paul and Stevenson, 1983; Waldron and Hunt, 1988; Wang and Waldron, 1987).

There are two types of singularities for a serial manipulator: *boundary singularity* and *interior singularity*. A boundary singularity occurs when the end effector is on the surface of the workspace boundary, and it usually happens when the manipulator is either in a fully stretched-out or a folded-back configuration. Boundary singularity can also occur when one of its actuators reaches its mechanical limit. An interior singularity occurs inside the workspace boundary. Several conditions may lead to an interior singularity. For example, when two or more joint axes line up on a straight line, the effects of a rotation about one joint axis can be canceled by counterrotation about another joint axis. Thus the end effector remains stationary even though the intermediate links of the manipulator may move in space. Another example of interior singularity occurs when four revolute joint axes are parallel to one another or intersect at a common point. For a manipulator of general geometry, the problem of identifying interior singularities becomes a much more complex problem. Basically, an interior singularity occurs whenever the screws of two or more joint axes become linearly dependent. Boundary singularities are not particularly serious, since they can always be avoided by arranging the tasks of manipulation far away from the workspace boundary. Interior singularity is more troublesome because it is more difficult to predict during the path planning process. The following examples illustrate the physical meaning of boundary and interior singularities.

4.11.1 Singular Configurations of a Planar 3-DOF Manipulator

Consider the planar 3-dof manipulator shown in Fig. 2.3 as an example. To identify the singular configurations, we equate the determinant of the conven-

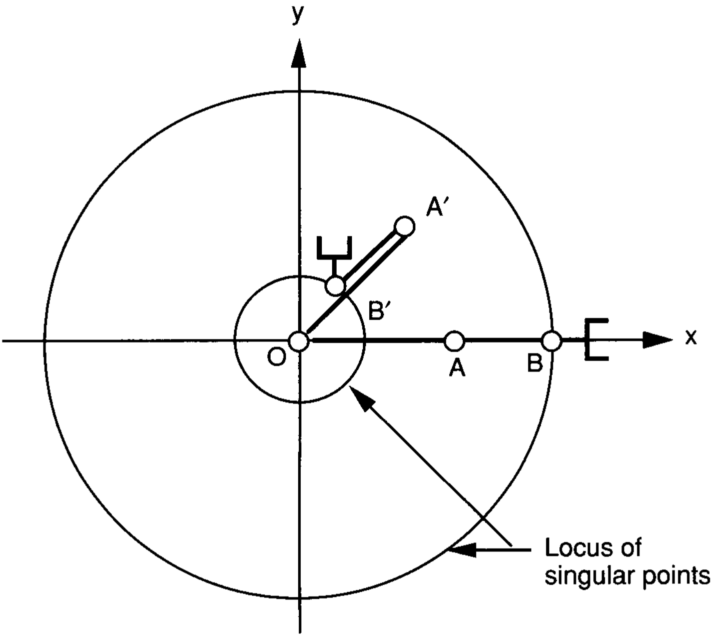

FIGURE 4.15. Locus of singular points.

tional Jacobian matrix, Eq. (4.67), to zero:

$$\det(J) = \begin{vmatrix} -(a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123}) & -(a_2 s\theta_{12} + a_3 s\theta_{123}) & -a_3 s\theta_{123} \\ (a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123}) & (a_2 c\theta_{12} + a_3 c\theta_{123}) & a_3 c\theta_{123} \\ 1 & 1 & 1 \end{vmatrix}$$

$$= a_1 a_2 s\theta_2 = 0. \tag{4.169}$$

We note that the joint angles θ_1 and θ_3 do not appear in Eq. (4.169) as predicted. Clearly, the manipulator assumes a singular configuration when $\theta_2 = 0$ or $\theta_2 = \pi$. The condition $\theta_2 = 0$ corresponds to a stretched-out configuration, and the condition $\theta_2 = \pi$ corresponds to a folded-back configuration, as depicted in Fig. 4.15. In either case, the manipulator loses 1 degree of freedom. The end effector can only move along the tangential direction of the arm; motion along the radial direction is not possible. This is a typical example of workspace boundary singularity.

4.11.2 Singular Configurations of the Stanford Manipulator

In this example we examine the singular configurations of the Stanford manipulator shown in Fig. 4.8. Equating the determinant of the screw-based Jacobian in Eq. (4.99) to zero, we obtain

$$\det({}^3J) = \begin{vmatrix} 0 & -1 & 0 & 0 & -s\theta_4 & c\theta_4 s\theta_5 \\ -s\theta_2 & 0 & 0 & 0 & c\theta_4 & s\theta_4 s\theta_5 \\ c\theta_2 & 0 & 0 & 1 & 0 & c\theta_5 \\ -d_3 s\theta_2 & 0 & 0 & 0 & 0 & 0 \\ -d_2 c\theta_2 & d_3 & 0 & 0 & 0 & 0 \\ -d_2 s\theta_2 & 0 & 1 & 0 & 0 & 0 \end{vmatrix} = -d_3^2 s\theta_2 s\theta_5 = 0. \quad (4.170)$$

The workspace boundary of the wrist center, P, is determined by the upper and lower limits of the sliding distance along the prismatic joint, namely the extreme values of d_3. Assuming that $d_3 \neq 0$, it can be concluded that the manipulator will lose (1) 1 degree of freedom if either $s\theta_2$ or $s\theta_5$ is equal to zero; (2) 2 degrees of freedom if both $s\theta_2$ and $s\theta_5$ are equal to zero simultaneously, and (3) 3 degrees of freedom, if condition 2 occurs at the workspace boundary.

The condition $s\theta_2 = 0$ is satisfied when $\theta_2 = 0$ or π. Under this condition, the arm points either vertically up or down, and the position of the wrist center is confined on a cylindrical surface of radius d_2. Hence it is a workspace boundary singularity.

The condition $s\theta_5 = 0$ is satisfied when $\theta_5 = 0$ or π. In this case the sixth joint axis is in line with the fourth. Therefore, any rotation about the fourth joint axis can be canceled by a counterrotation about the sixth joint axis. That is, the wrist can perform a *self-motion* with no effect on the orientation of the end effector. This is an interior singularity.

Mathematically, another singularity occurs at $d_3 = 0$. When $d_3 = 0$, any rotation about the second joint axis (z_1) has no effects on the linear velocity of the wrist center. However, in practice, the minimum value of d_3 is determined by the physical construction of the robot arm.

4.11.3 Singular Configurations of the Elbow Manipulator

In this example we examine the singular configurations of the elbow manipulator shown in Fig. 4.10. Equating the determinant of the screw-based Jacobian given by Eq. (4.115) to zero, we obtain

$$\det({}^4J) = \begin{vmatrix} s\theta_{234} & 0 & 0 & 0 & 0 & s\theta_5 \\ 0 & -1 & -1 & -1 & 0 & -c\theta_5 \\ c\theta_{234} & 0 & 0 & 0 & 1 & 0 \\ 0 & a_2 s\theta_{34} + a_3 s\theta_4 & a_3 s\theta_4 & 0 & 0 & 0 \\ x_{51} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_2 c\theta_{34} + a_3 c\theta_4 + a_4 & a_3 c\theta_4 + a_4 & a_4 & 0 & 0 \end{vmatrix}$$

$$= -a_2 a_3 x_{51} s\theta_3 s\theta_5 = 0, \quad (4.171)$$

where $x_{51} = a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234}$.

Assuming that $a_2 \neq 0$ and $a_3 \neq 0$, it can be concluded that singularities occur when at least one of the three factors $s\theta_3$, $s\theta_5$, and x_{51} is equal to zero. Specifically, the manipulator will lose (1) 1 degree of freedom if either $s\theta_3$, $s\theta_5$, or x_{51} is equal to zero; (2) 2 degrees of freedom if any two of $s\theta_3$, $s\theta_5$, and x_{51} are equal to zero simultaneously, and (3) 3 degrees of freedom if $s\theta_3$, $s\theta_5$, and x_{51} are all equal to zero. The manipulator cannot lose more than 3 degrees of freedom.

The condition $s\theta_3 = 0$ is satisfied when $\theta_3 = 0$ or π. Under this condition, the second and third links are either in a stretched-out or folded-back configuration. This can be considered as a workspace boundary singularity from a regional structure point of view.

The condition $s\theta_5 = 0$ is satisfied when $\theta_5 = 0$ or π. Under such a condition, the sixth joint axis becomes parallel to the second, third, and fourth joint axes, and the manipulator loses 1 degree of freedom. This type of singularity can occur within the workspace boundary and therefore is an interior singularity.

The condition $x_{51} = a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234} = 0$ is satisfied when the wrist center, O_5, is brought back to the origin of the first link frame, (x_1, y_1, z_1). Thus any rotation about the second joint axis has no effect on the position of the wrist center. Since the first and second joint axes intersect each other, any rotation about the first joint axis has no effect on the position of the wrist center either.

4.12 SUMMARY

In this chapter, the differential kinematics of serial manipulators was studied. First, the differential kinematics of a rigid body was reviewed. The differential matrices of transformation and the concept of instantaneous screw motion were introduced for the kinematic analysis of serial manipulators. Then the manipulator Jacobian matrix was defined. Both the conventional method and the method of instantaneous screw motions were described. A recursive method for derivation of the screw-based Jacobian matrix was presented. It was shown that the Jacobian of a manipulator can be greatly simplified by expressing the screws in a properly chosen reference frame. To facilitate the inverse velocity analysis associated with the trajectory planning problem, a 6×6 transformation matrix was introduced. Finally, the singularity of a robot manipulator was defined. It was shown that the singularity of a serial manipulator is independent of the first and last joint variables. Several serial manipulators were studied to illustrate the principle.

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EXERCISES

1. Let ${}^A\mathbf{v}_p$ and ${}^A\mathbf{v}_q$ be the velocities of two points in a rigid body B that is moving with respect to a fixed frame A. Derive the angular velocity of B in terms of ${}^A\mathbf{v}_p$ and ${}^A\mathbf{v}_q$.

2. Consider the spatial 3-dof, $3R$ manipulator shown in Fig. 2.19. Calculate the velocity of the point Q in the end effector as a function of joint rates. Express it in the fixed frame.

3. Figure 4.16 shows a planar 3-dof, $3R$ manipulator. We wish to move the end effector, Q, along the x-axis at 1.0 m/s and, at the same time, keep the direction of approach, the x_3-axis, in the $-y_0$-direction. Calculate the joint rates required to accomplish this task. Under what conditions will the joint rates approach infinity?

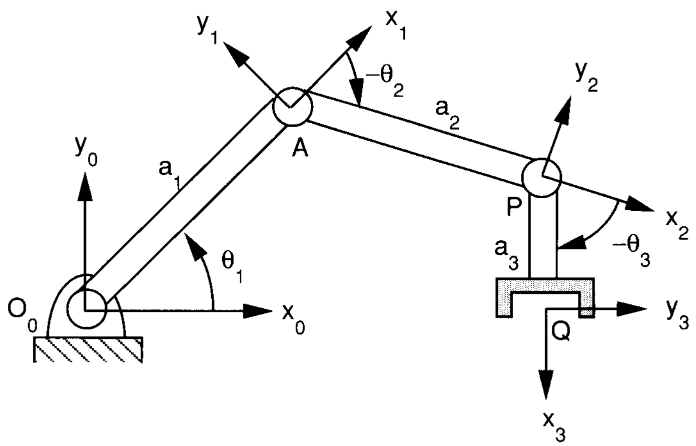

FIGURE 4.16. Planar 3-dof manipulator.

4. For the SCARA robot shown in Fig. 2.4, derive the velocity of point Q and the angular velocity of the end effector in terms of the joint rates. What is the appropriate form of a Jacobian matrix? Express it in the fixed frame.

5. For the 5-dof Scorbot robot shown in Fig. 2.5, derive the velocity of point Q and the angular velocity of the end effector in terms of the joint rates. Formulate the conventional Jacobian matrix. Is this Jacobian a square matrix?

6. Derive the conventional Jacobian matrix for the elbow manipulator shown in Fig. 4.10.

7. Calculate the screw-based Jacobian matrix of the SCARA robot shown in Fig. 2.4. Express it in the (x_3, y_3, z_3) coordinate frame.

8. Derive the screw-based Jacobian matrix for the 6-dof Fanuc S-900W robot shown in Fig. 2.8. Express it in the (x_3, y_3, z_3) coordinate frame.

9. For the spatial 3-dof robot shown in Fig. 2.19, identify a set of joint angles for which the manipulator is at a workspace boundary singularity, and another set of joint angles for which the manipulator is at a workspace interior singularity.

10. Show that a spatial manipulator with three revolute joints and nonzero link lengths always contains a locus of workspace interior singular points.

11. What are the necessary conditions for a spatial manipulator with three revolute joints and nonzero link lengths to possess isotropic points?

12. Find the singular loci associated with the SCARA robot shown in Fig. 2.4.

13. Derive the singular points associated with the 5-dof Scorbot robot shown in Fig. 2.5. Explain whether they are workspace boundary or interior singularities.